

of a 0.02 per cent V/V solution of *toluene R* in *hexane R* or *heptane R*, with respectively *hexane R* or *heptane R* as the compensation liquid.

Acceptance criterion

For measurements taken with a solution prepared as described above, the minimum ratio of the absorbance at the maximum (269 nm) to that at the minimum (266 nm) is stated in the monograph.

SYSTEM SUITABILITY

System suitability tests may be required prior to sample measurement to verify critical parameters that may have an impact on the result.

These tests may cover wavelength accuracy, absorbance accuracy, stray light and photometric linearity. System functionality tests, for example those performed as part of equipment autotesting, may be considered part of the system suitability tests.

In the case of UV-Vis detection for chromatographic systems, additional system suitability tests are applicable if prescribed in the monograph and/or in general chapter 2.2.46. *Chromatographic separation techniques*.



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2.2.26. PAPER CHROMATOGRAPHY

ASCENDING PAPER CHROMATOGRAPHY

Apparatus. The apparatus consists of a glass tank of suitable size for the chromatographic paper used, ground at the top to take a closely fitting lid. In the top of the tank is a device which suspends the chromatographic paper and is capable of being lowered without opening the chamber. In the bottom of the tank is a dish to contain the mobile phase into which the paper may be lowered. The chromatographic paper consists of suitable filter paper, cut into strips of sufficient length and not less than 2.5 cm wide; the paper is cut so that the mobile phase runs in the direction of the grain of the paper.

Method. Place in the dish a layer 2.5 cm deep of the mobile phase prescribed in the monograph. If prescribed in the monograph, pour the stationary phase between the walls of the tank and the dish. Close the tank and allow to stand for 24 h at 20 °C to 25 °C. Maintain the tank at this temperature throughout the subsequent procedure. Draw a fine pencil line horizontally across the paper 3 cm from one end. Using a micro pipette, apply to a spot on the pencil line the volume of the solution prescribed in the monograph. If the total volume to be applied would produce a spot more than 10 mm in diameter, apply the solution in portions allowing each to dry before the next application. When more than one chromatogram is to be run on the same strip of paper, space the solutions along the pencil line at points not less than 3 cm apart. Insert the paper into the tank, close the lid and allow to stand for 1 h 30 min. Lower the paper into the mobile phase and allow elution to proceed for the prescribed distance or time. Remove the paper from the tank and allow to dry in air. Protect the paper from bright light during the elution process.

DESCENDING PAPER CHROMATOGRAPHY

Apparatus. The apparatus consists of a glass tank of suitable size for the chromatographic paper used, ground at the top to take a closely fitting glass lid. The lid has a central hole about 1.5 cm in diameter closed by a heavy glass plate or a stopper. In the upper part of the tank is suspended a solvent trough with a device for holding the chromatographic paper. On each side of the trough, parallel to and slightly above its upper edges, are two glass guide rods to support the paper in such a manner that no part of it is in contact with the

walls of the tank. The chromatographic paper consists of suitable filter paper, cut into strips of sufficient length, and of any convenient width between 2.5 cm and the length of the trough; the paper is cut so that the mobile phase runs in the direction of the grain of the paper.

Method. Place in the bottom of the tank a layer 2.5 cm deep of the solvent prescribed in the monograph, close the tank and allow to stand for 24 h at 20 °C to 25 °C. Maintain the tank at this temperature throughout the subsequent procedure. Draw a fine pencil line horizontally across the paper at such a distance from one end that when this end is secured in the solvent trough and the remainder of the paper is hanging freely over the guide rod, the line is a few centimetres below the guide rod and parallel with it. Using a micro-pipette, apply on the pencil line the volume of the solution prescribed in the monograph. If the total volume to be applied would produce a spot more than 10 mm in diameter, apply the solution in portions, allowing each to dry before the next application. When more than one chromatogram is to be run on the same strip of paper, space the solutions along the pencil line at points not less than 3 cm apart. Insert the paper in the tank, close the lid, and allow to stand for 1 h 30 min. Introduce into the solvent trough, through the hole in the lid, a sufficient quantity of the mobile phase, close the tank and allow elution to proceed for the prescribed distance or time. Remove the paper from the tank and allow to dry in air. The paper should be protected from bright light during the elution process.

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2.2.27. THIN-LAYER CHROMATOGRAPHY

Thin-layer chromatography is a separation technique in which a stationary phase consisting of an appropriate material is spread in a uniform thin layer on a support (plate) of glass, metal or plastic. Solutions of analytes are deposited on the plate prior to development. The separation is based on adsorption, partition, ion-exchange or on combinations of these mechanisms and is carried out by migration (development) of solutes (solutions of analytes) in a solvent or a suitable mixture of solvents (mobile phase) through the thin-layer (stationary phase).

APPARATUS

Plates. The chromatography is carried out using pre-coated plates as described under *Reagents* (4.1.1). The particle size of the silica gel is indicated after the name of the reagent in the tests where it is used.

Pre-treatment of the plates. It may be necessary to wash the plates prior to separation. This can be done by migration of an appropriate solvent. The plates may also be impregnated by procedures such as development, immersion or spraying. At the time of use, the plates may be activated, if necessary, by heating in an oven at 120 °C for 20 min.

Chromatographic tank with a flat bottom or twin trough, of inert, transparent material, of a size suitable for the plates used and provided with a tightly fitting lid. For horizontal development the tank is provided with a trough for the mobile phase and it additionally contains a device for directing the mobile phase to the stationary phase.

Micropipettes, microsyringes, calibrated disposable capillaries or other application devices suitable for the proper application of the solutions.

Fluorescence detection device to measure direct fluorescence or the inhibition of fluorescence.



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2.2.27. THIN-LAYER CHROMATOGRAPHY

PRINCIPLE

Thin-layer chromatography is a separation technique in which a stationary phase consisting of an appropriate material is spread in a uniform thin layer on a support (plate) of glass, metal or plastic. Solutions of analytes are deposited on the plate prior to development. The separation is based on adsorption, partition, ion-exchange or combinations of these mechanisms and is carried out by migration (development) of solutes (solutions of analytes) in a solvent or a suitable mixture of solvents (mobile phase) through the thin layer (stationary phase).

EQUIPMENT

Plates. The chromatography is carried out using pre-coated plates as described under *Reagents* (4.1.1). In a monograph in which both normal and high-performance plates may be used, the particle size of the silica gel is indicated in parentheses () after the name of the reagent.

Pre-treatment of the plates. It may be necessary to wash the plates prior to separation. This can be done by migration of an appropriate solvent. The plates may also be impregnated by procedures such as development, immersion or spraying. At the time of use, the plates may be activated, if necessary, by heating in an oven at 120 °C for 20 min.

Chromatographic tank with a flat bottom or twin trough, of inert, transparent material, of a size suitable for the plates used and provided with a tightly fitting lid. For horizontal development the tank is provided with a trough for the mobile phase and it additionally contains a device for directing the mobile phase to the stationary phase.

Micropipettes, microsyringes, calibrated disposable capillaries or other application devices suitable for the proper application of the solutions.

Fluorescence detection device to measure direct fluorescence or the inhibition of fluorescence.

Visualisation devices and reagents. Suitable devices are used for derivatisation to transfer to the plate reagents by spraying, immersion or exposure to vapour and, where applicable, to facilitate heating for visualisation of separated components.

Documentation. A device may be used to provide documentation of the visualised chromatogram, for example a photograph or a computer file.

PROCEDURE

Sample application. Apply the prescribed volume of the solutions at a suitable distance from the lower edge and from the sides of the plate and on a line parallel to the lower edge; allow an interval of at least 10 mm (5 mm on high-performance plates) between the centres of circular spots and 5 mm (2 mm on high-performance plates) between the edges of bands. Apply the solutions in sufficiently small portions to obtain circular spots 2-5 mm in diameter (1-2 mm on high-performance plates) or bands 10-20 mm (5-10 mm on high-performance plates) by 1-2 mm.

In a monograph in which both normal and high-performance plates may be used, the working conditions for high-performance plates are given in brackets [] after those for normal plates.

Vertical development. Line the walls of the chromatographic tank with filter paper. Pour into the chromatographic tank a sufficient quantity of the mobile phase for the size of the

tank to give after impregnation of the filter paper a layer of appropriate depth related to the dimension of the plate to be used. For saturation of the chromatographic tank, replace the lid and allow to stand at 20-25 °C for 1 h. Unless otherwise indicated in the monograph, the chromatographic separation is performed in a saturated tank. Apply the prescribed volume of solutions as described above. When the solvent has evaporated from the applied solutions, place the plate in the chromatographic tank, ensuring that the plate is as vertical as possible and that the spots or bands are above the surface of the mobile phase. Close the chromatographic tank, maintain it at 20-25 °C and protect from sunlight. Remove the plate when the mobile phase has moved over the prescribed distance, measured between the points of application and the solvent front. Dry the plate and visualise the chromatograms as prescribed.

For two-dimensional chromatography, dry the plates after the first development and carry out a second development in a direction perpendicular to that of the first development.

Horizontal development. Apply the prescribed volume of the solutions as described above. When the solvent has evaporated from the applied solutions, introduce a sufficient quantity of the mobile phase into the trough of the chamber using a syringe or pipette, place the plate in the chamber after verifying that the latter is horizontal and connect the mobile phase direction device according to the manufacturer's instructions. If prescribed, develop the plate starting simultaneously at both ends. Close the chamber and maintain it at 20-25 °C. Remove the plate when the mobile phase has moved over the distance prescribed in the monograph. Dry the plate and visualise the chromatograms as prescribed.

For two-dimensional chromatography, dry the plates after the first development and carry out a second development in a direction perpendicular to that of the first development.

Parameters for assessing the suitability of the system are described in general chapter 2.2.46. *Chromatographic separation techniques*. The extent to which adjustments of the chromatographic conditions can be made are also given in that general chapter.

VISUAL EVALUATION

Identification. The principal spot in the chromatogram obtained with the test solution is visually compared to the corresponding spot in the chromatogram obtained with the reference solution by comparing the colour, the size and the retardation factor (R_f) of both spots.

The retardation factor (R_f) is defined as the ratio of the distance from the point of application to the centre of the spot and the distance travelled by the solvent front from the point of application.

Verification of the separating power for identification. The performance of the plate, given by the test for chromatographic separation in the description of the corresponding reagent in general chapter 4.1.1. *Reagents*, is usually sufficient. An additional performance criterion may be prescribed in the individual monograph to control the system suitability.

Related substances test. The secondary spot(s) in the chromatogram obtained with the test solution is (are) visually compared to either the corresponding spot(s) in the chromatogram obtained with the reference solution containing the impurity(ies) or the spot in the chromatogram obtained with the reference solution prepared from a dilution of the test solution.

Verification of the separating power. The requirements for the verification of the separating power are prescribed in the monographs concerned.

Verification of the detecting power. The detecting power is satisfactory if a spot or band is clearly visible in the chromatogram obtained with the most dilute reference solution.

QUANTITATIVE MEASUREMENT

The requirements for resolution and separation are prescribed in the monographs concerned.

Substances separated by thin-layer chromatography and responding to UV-Vis irradiation can be determined directly on the plate, using appropriate instrumentation. While moving the plate or the measuring device, examine the plate by measuring the reflectance of the incident light. Similarly, fluorescence may be measured using an appropriate optical system. Substances containing radionuclides can be quantified in 3 ways: either directly by moving the plate alongside a suitable counter or vice versa (see *Radiopharmaceutical preparations (0125)*), by cutting the plates into strips and measuring the radioactivity on each individual strip using a suitable counter or by scraping off the stationary phase, dissolving it in a suitable scintillation cocktail and measuring the radioactivity using a liquid scintillation counter.

Equipment. The equipment for direct measurement on the plate consists of:

- a device for exact positioning and reproducible dispensing of the amount of substances onto the plate;
- a mechanical device to move the plate or the measuring device along the *x*-axis or the *y*-axis;
- a recorder and a suitable integrator or a computer;
- *for substances responding to UV-Vis irradiation:* a photometer with a source of light, an optical device able to generate monochromatic light and a photo cell of adequate sensitivity, for the measurement of reflectance or transmittance; if fluorescence is measured, a suitable filter is required to prevent light used for excitation from reaching the detector while permitting emitted light or a specific portion thereof to pass;
- *for substances containing radionuclides:* a suitable counter for radioactivity; the linearity range of the counting device is to be verified.

Procedure. Prepare the solution of the substance to be examined (test solution) as prescribed in the monograph and, if necessary, prepare the reference solutions of the substance to be determined using the same solvent as in the test solution. Apply the same volume of each solution to the plate and develop.

Substances responding to UV-Vis irradiation. Prepare and apply not fewer than 3 reference solutions of the substance to be examined, the concentrations of which span the expected value in the test solution (about 80 per cent, 100 per cent and 120 per cent). Treat with the prescribed reagent, if necessary, and record the reflectance, transmittance or fluorescence in the chromatograms obtained with the test and reference solutions. Use the measured results for the calculation of the amount of substance in the test solution.

Substances containing radionuclides. Prepare and apply a test solution containing about 100 per cent of the expected value. Determine the radioactivity as a function of the path length and report the radioactivity in each resulting peak as a percentage of the total amount of radioactivity.

is a carrier gas moving through or passing the stationary phase contained in a column. It is applicable to substances or their derivatives that are volatilised under the temperatures employed.

GC is mainly based on mechanisms of adsorption or mass distribution.

EQUIPMENT

The equipment typically consists of:

- an injector;
- a chromatographic column contained in an oven;
- one or more detector(s);
- a data acquisition system.

The carrier gas flows through the column and then through the detector at a controlled rate or pressure.

The chromatography is carried out either at a constant temperature or according to a given temperature programme.

INJECTORS

Injection may be carried out either into a vaporisation chamber, which may be equipped with a stream splitter, or directly at the head of the column using a syringe or an injection valve.

Injections of vapour phase may be effected by static or dynamic head-space injection systems.

Dynamic head-space (purge and trap) injection systems include a sparging device by which volatile substances in solution are swept into an adsorbent column maintained at a low temperature. Retained substances are then desorbed into the mobile phase by rapid heating of the adsorbent column.

Static head-space injection systems include a thermostatically controlled sample heating chamber in which closed vials containing solid or liquid samples are placed for a fixed period of time to allow equilibration of the volatile components of the sample between the non-gaseous phase and the vapour phase. After equilibration, a predetermined amount of the head-space of the vial is flushed into the gas chromatograph.

STATIONARY PHASES

Stationary phases are contained in columns that may be:

- a capillary column whose stationary phase may be a solid coating the inner surface of the column (e.g. macrogol 20 000) or a liquid deposited on the inner surface (e.g. dimethylpolysiloxane); in the latter case it may be chemically bonded to the inner surface;
- a column packed with the stationary phase that may be a solid phase (e.g. alumina, silica) or an inert solid support (usually a porous polymer) impregnated or coated with a liquid.

Capillary columns, made of fused silica, are 0.1 mm to 0.53 mm in internal diameter ($\varnothing$) and at least 5 m in length. The stationary phase is a film 0.1 μ m to 5.0 μ m thick.

Packed columns, made of glass or metal, are usually 1 m to 3 m in length with an internal diameter ($\varnothing$) of 2 mm to 4 mm.

MOBILE PHASES

Retention time and peak efficiency depend on the carrier gas flow rate; retention time is directly proportional to column length and resolution is proportional to the square root of the column length.

The carrier gas flow rate is usually expressed in millilitres per minute at atmospheric pressure and at the stated temperature. Flow rate is measured at the detector outlet, either with a calibrated mechanical device or with a bubble tube, while the column is at operating temperature.

The linear velocity of the carrier gas through a column is inversely proportional to the square of the internal diameter of the column for a given flow volume.

Helium, nitrogen and hydrogen are commonly used carrier gases.



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2.2.28. GAS CHROMATOGRAPHY

PRINCIPLE

Gas chromatography (GC) is a chromatographic separation technique based on the difference in the distribution of species between two non-miscible phases in which the mobile phase

QUANTITATIVE MEASUREMENT

The requirements for resolution and separation are prescribed in the monographs concerned.

Substances separated by thin-layer chromatography and responding to UV-Vis irradiation can be determined directly on the plate, using appropriate instrumentation. While moving the plate or the measuring device, examine the plate by measuring the reflectance of the incident light. Similarly, fluorescence may be measured using an appropriate optical system. Substances containing radionuclides can be quantified in 3 ways: either directly by moving the plate alongside a suitable counter or vice versa (see *Radiopharmaceutical preparations (0125)*), by cutting the plates into strips and measuring the radioactivity on each individual strip using a suitable counter or by scraping off the stationary phase, dissolving it in a suitable scintillation cocktail and measuring the radioactivity using a liquid scintillation counter.

Equipment. The equipment for direct measurement on the plate consists of:

- a device for exact positioning and reproducible dispensing of the amount of substances onto the plate;
- a mechanical device to move the plate or the measuring device along the *x*-axis or the *y*-axis;
- a recorder and a suitable integrator or a computer;
- *for substances responding to UV-Vis irradiation:* a photometer with a source of light, an optical device able to generate monochromatic light and a photo cell of adequate sensitivity, for the measurement of reflectance or transmittance; if fluorescence is measured, a suitable filter is required to prevent light used for excitation from reaching the detector while permitting emitted light or a specific portion thereof to pass;
- *for substances containing radionuclides:* a suitable counter for radioactivity; the linearity range of the counting device is to be verified.

Procedure. Prepare the solution of the substance to be examined (test solution) as prescribed in the monograph and, if necessary, prepare the reference solutions of the substance to be determined using the same solvent as in the test solution. Apply the same volume of each solution to the plate and develop.

Substances responding to UV-Vis irradiation. Prepare and apply not fewer than 3 reference solutions of the substance to be examined, the concentrations of which span the expected value in the test solution (about 80 per cent, 100 per cent and 120 per cent). Treat with the prescribed reagent, if necessary, and record the reflectance, transmittance or fluorescence in the chromatograms obtained with the test and reference solutions. Use the measured results for the calculation of the amount of substance in the test solution.

Substances containing radionuclides. Prepare and apply a test solution containing about 100 per cent of the expected value. Determine the radioactivity as a function of the path length and report the radioactivity in each resulting peak as a percentage of the total amount of radioactivity.

is a carrier gas moving through or passing the stationary phase contained in a column. It is applicable to substances or their derivatives that are volatilised under the temperatures employed.

GC is mainly based on mechanisms of adsorption or mass distribution.

EQUIPMENT

The equipment typically consists of:

- an injector;
- a chromatographic column contained in an oven;
- one or more detector(s);
- a data acquisition system.

The carrier gas flows through the column and then through the detector at a controlled rate or pressure.

The chromatography is carried out either at a constant temperature or according to a given temperature programme.

INJECTORS

Injection may be carried out either into a vaporisation chamber, which may be equipped with a stream splitter, or directly at the head of the column using a syringe or an injection valve.

Injections of vapour phase may be effected by static or dynamic head-space injection systems.

Dynamic head-space (purge and trap) injection systems include a sparging device by which volatile substances in solution are swept into an adsorbent column maintained at a low temperature. Retained substances are then desorbed into the mobile phase by rapid heating of the adsorbent column.

Static head-space injection systems include a thermostatically controlled sample heating chamber in which closed vials containing solid or liquid samples are placed for a fixed period of time to allow equilibration of the volatile components of the sample between the non-gaseous phase and the vapour phase. After equilibration, a predetermined amount of the head-space of the vial is flushed into the gas chromatograph.

STATIONARY PHASES

Stationary phases are contained in columns that may be:

- a capillary column whose stationary phase may be a solid coating the inner surface of the column (e.g. macrogol 20 000) or a liquid deposited on the inner surface (e.g. dimethylpolysiloxane); in the latter case it may be chemically bonded to the inner surface;
- a column packed with the stationary phase that may be a solid phase (e.g. alumina, silica) or an inert solid support (usually a porous polymer) impregnated or coated with a liquid.

Capillary columns, made of fused silica, are 0.1 mm to 0.53 mm in internal diameter ($\varnothing$) and at least 5 m in length. The stationary phase is a film 0.1 μ m to 5.0 μ m thick.

Packed columns, made of glass or metal, are usually 1 m to 3 m in length with an internal diameter ($\varnothing$) of 2 mm to 4 mm.

MOBILE PHASES

Retention time and peak efficiency depend on the carrier gas flow rate; retention time is directly proportional to column length and resolution is proportional to the square root of the column length.

The carrier gas flow rate is usually expressed in millilitres per minute at atmospheric pressure and at the stated temperature. Flow rate is measured at the detector outlet, either with a calibrated mechanical device or with a bubble tube, while the column is at operating temperature.

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Helium, nitrogen and hydrogen are commonly used carrier gases.



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DETECTORS

Flame-ionisation detectors are usually employed but other detectors such as electron-capture, nitrogen-phosphorus, mass spectrometric, thermal conductivity or infrared spectrophotometric detectors may also be used.

PROCEDURE

Equilibrate the column, the injector and the detector at the temperatures and the gas flow rates/pressures specified in the monograph until a stable baseline is achieved. Prepare the test solution(s) and the reference solution(s) as prescribed. The solutions injected must be free from solid particles.

Criteria for assessing the suitability of the system are described in general chapter 2.2.46. *Chromatographic separation techniques*. The extent to which adjustments of chromatographic conditions can be made are also given in that general chapter.

Static head-space gas chromatography

Static head-space gas chromatography is a technique particularly suitable for separating and determining volatile components present in solid or liquid samples. The technique is based on the analysis of the vapour phase in equilibrium with the solid or liquid phase.

EQUIPMENT

The equipment consists of a gas chromatograph provided with a sample-introduction device that may be connected to a module that automatically controls the pressure and the temperature. If necessary, a device for eliminating solvents can be added.

The sample to be analysed is introduced into a container fitted with a suitable stopper and a valve-system that permits the passage of the carrier gas. The container is placed in a thermostatically controlled chamber at a temperature set according to the substance to be examined.

The sample is held at this temperature long enough to allow equilibration between the solid or liquid phase and the vapour phase.

The carrier gas is introduced into the container and, after the prescribed time, a suitable valve is opened so that the gas expands towards the chromatographic column, taking the volatilised components with it.

Instead of using a chromatograph specifically equipped for the introduction of samples, it is also possible to use airtight syringes and a conventional chromatograph. Equilibration is then carried out in a separate chamber and the vapour phase is carried onto the column, while necessary precautions are taken to avoid any changes in the equilibrium.

PROCEDURE

Using the reference solutions, determine suitable instrument settings to produce an adequate response.

DIRECT CALIBRATION

Introduce into separate, identical containers the test solution and each of the reference solutions, as prescribed in the monograph, avoiding contact between the sampling device and the samples.

Close the containers hermetically and place in the thermostatically controlled chamber set to the temperature and pressure prescribed in the monograph; after equilibration, carry out the chromatography under the prescribed conditions.

STANDARD ADDITIONS

To a set of identical suitable containers, add equal volumes of the test solution. To all but one of the containers, add suitable quantities of a reference solution containing a known concentration of the substance to be determined so as to produce a series of solutions containing steadily increasing concentrations of the substance.

Close the containers hermetically and place in the thermostatically controlled chamber set to the temperature and pressure prescribed in the monograph; after equilibration, carry out the chromatography under the prescribed conditions. Calculate the linear equation of the graph using a least-squares fit, and derive from it the concentration of the substance to be determined in the test solution.

Alternatively, plot on a graph the mean of readings against the added quantity of the substance to be determined. Extrapolate the line joining the points on the graph until it meets the concentration axis. The distance between this point and the intersection of the axes represents the concentration of the substance to be determined in the test solution.

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**2.2.29. LIQUID CHROMATOGRAPHY****PRINCIPLE**

Liquid chromatography (LC) is a chromatographic separation technique based on the difference in the distribution of species between 2 non-miscible phases, in which the mobile phase is a liquid that percolates through a stationary phase contained in a column.

LC is mainly based on mechanisms of adsorption, mass distribution, ion exchange, size exclusion or stereochemical interaction.

Unless otherwise specified, all the information below is valid for both standard LC and LC using reduced particle-size columns (e.g. sub-2.0 µm).

The latter requires instrumentation that is able to withstand higher pressures (typically up to 100 MPa, i.e. about 15 000 psi), generates lower extra-column band broadening, provides improved gradient mixing and allows a higher sampling rate in the detection system.

EQUIPMENT

The equipment typically consists of:

- a pumping system;
- an injector;
- a chromatographic column (a column temperature controller may be used);
- one or more detectors;
- a data acquisition system.

The mobile phase is supplied from one or more reservoirs and is pumped to the injector, then through the column, usually at a constant rate, and then through the detector(s).

PUMPING SYSTEMS

LC pumping systems deliver the mobile phase at a controlled flow rate. Pressure fluctuations are to be minimised, for example by passing the pressurised solvent through a pulse-dampening device. Tubing and connections are capable of withstanding the pressures developed by the pumping system. LC pumps may be fitted with a facility for 'bleeding' the system of entrapped air bubbles.

Microprocessor-controlled pumping systems are capable of accurately delivering a mobile phase of either constant (isocratic elution) or varying (gradient elution) composition, according to a defined programme. In the case of gradient elution, pumping systems that deliver solvent(s) from several reservoirs are available and solvent mixing can be achieved on either the low- or the high-pressure side of the pump(s).

INJECTORS

The sample solution is introduced into the flowing mobile phase at or near the head of the column using an injection system that can operate at high pressure. Fixed-loop

DETECTORS

Flame-ionisation detectors are usually employed but other detectors such as electron-capture, nitrogen-phosphorus, mass spectrometric, thermal conductivity or infrared spectrophotometric detectors may also be used.

PROCEDURE

Equilibrate the column, the injector and the detector at the temperatures and the gas flow rates/pressures specified in the monograph until a stable baseline is achieved. Prepare the test solution(s) and the reference solution(s) as prescribed. The solutions injected must be free from solid particles.

Criteria for assessing the suitability of the system are described in general chapter 2.2.46. *Chromatographic separation techniques*. The extent to which adjustments of chromatographic conditions can be made are also given in that general chapter.

Static head-space gas chromatography

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EQUIPMENT

The equipment consists of a gas chromatograph provided with a sample-introduction device that may be connected to a module that automatically controls the pressure and the temperature. If necessary, a device for eliminating solvents can be added.

The sample to be analysed is introduced into a container fitted with a suitable stopper and a valve-system that permits the passage of the carrier gas. The container is placed in a thermostatically controlled chamber at a temperature set according to the substance to be examined.

The sample is held at this temperature long enough to allow equilibration between the solid or liquid phase and the vapour phase.

The carrier gas is introduced into the container and, after the prescribed time, a suitable valve is opened so that the gas expands towards the chromatographic column, taking the volatilised components with it.

Instead of using a chromatograph specifically equipped for the introduction of samples, it is also possible to use airtight syringes and a conventional chromatograph. Equilibration is then carried out in a separate chamber and the vapour phase is carried onto the column, while necessary precautions are taken to avoid any changes in the equilibrium.

PROCEDURE

Using the reference solutions, determine suitable instrument settings to produce an adequate response.

DIRECT CALIBRATION

Introduce into separate, identical containers the test solution and each of the reference solutions, as prescribed in the monograph, avoiding contact between the sampling device and the samples.

Close the containers hermetically and place in the thermostatically controlled chamber set to the temperature and pressure prescribed in the monograph; after equilibration, carry out the chromatography under the prescribed conditions.

STANDARD ADDITIONS

To a set of identical suitable containers, add equal volumes of the test solution. To all but one of the containers, add suitable quantities of a reference solution containing a known concentration of the substance to be determined so as to produce a series of solutions containing steadily increasing concentrations of the substance.

Close the containers hermetically and place in the thermostatically controlled chamber set to the temperature and pressure prescribed in the monograph; after equilibration, carry out the chromatography under the prescribed conditions. Calculate the linear equation of the graph using a least-squares fit, and derive from it the concentration of the substance to be determined in the test solution.

Alternatively, plot on a graph the mean of readings against the added quantity of the substance to be determined. Extrapolate the line joining the points on the graph until it meets the concentration axis. The distance between this point and the intersection of the axes represents the concentration of the substance to be determined in the test solution.

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LC is mainly based on mechanisms of adsorption, mass distribution, ion exchange, size exclusion or stereochemical interaction.

Unless otherwise specified, all the information below is valid for both standard LC and LC using reduced particle-size columns (e.g. sub-2.0 µm).

The latter requires instrumentation that is able to withstand higher pressures (typically up to 100 MPa, i.e. about 15 000 psi), generates lower extra-column band broadening, provides improved gradient mixing and allows a higher sampling rate in the detection system.

EQUIPMENT

The equipment typically consists of:

- a pumping system;
- an injector;
- a chromatographic column (a column temperature controller may be used);
- one or more detectors;
- a data acquisition system.

The mobile phase is supplied from one or more reservoirs and is pumped to the injector, then through the column, usually at a constant rate, and then through the detector(s).

PUMPING SYSTEMS

LC pumping systems deliver the mobile phase at a controlled flow rate. Pressure fluctuations are to be minimised, for example by passing the pressurised solvent through a pulse-dampening device. Tubing and connections are capable of withstanding the pressures developed by the pumping system. LC pumps may be fitted with a facility for 'bleeding' the system of entrapped air bubbles.

Microprocessor-controlled pumping systems are capable of accurately delivering a mobile phase of either constant (isocratic elution) or varying (gradient elution) composition, according to a defined programme. In the case of gradient elution, pumping systems that deliver solvent(s) from several reservoirs are available and solvent mixing can be achieved on either the low- or the high-pressure side of the pump(s).

INJECTORS

The sample solution is introduced into the flowing mobile phase at or near the head of the column using an injection system that can operate at high pressure. Fixed-loop

and variable-volume devices operated manually or by an autosampler are used. Partial filling of loops during manual injection may adversely affect injection volume precision.

STATIONARY PHASES

There are many types of stationary phases employed in LC, including:

- silica or alumina, commonly used in normal-phase LC (polar stationary phase and non-polar mobile phase), where the separation is based on differences in adsorption on the stationary phase and/or mass distribution between the mobile phase and the stationary phase (partition chromatography);
- a variety of chemically modified supports prepared from polymers, silica or porous graphite, used in normal-phase and reversed-phase LC (non-polar stationary phase and polar mobile phase), where the separation is based principally on partition of the molecules;
- resins or polymers with acidic or basic groups, used in ion-exchange chromatography, where separation is based on competition between the ions to be separated and those in the mobile phase;
- porous silica or polymers, used in size-exclusion chromatography (2.2.30), where separation is based on differences between the volumes of the molecules, corresponding to steric exclusion;
- specially modified stationary phases, e.g. cellulose or amylose derivatives, proteins or peptides, cyclodextrins etc., for the separation of enantiomers (chiral chromatography).

Most separations are based on reversed-phase LC utilising chemically modified silica as the stationary phase. The surface of the support, i.e. the silanol groups of silica, is reacted with various silane reagents to produce covalently bound silyl derivatives covering a varying number of active sites on the surface of the support. The nature of the bonded phase is an important parameter for determining the separation properties of the chromatographic system.

Unless otherwise stated by the manufacturer, silica-based reversed-phase columns are considered to be stable in mobile phases having an apparent pH in the range 2.0 to 8.0. Columns containing porous graphite or particles of polymeric materials such as styrene-divinylbenzene copolymer are stable over a wider pH range.

Analysis using normal-phase LC with unmodified silica or polar, chemically modified silica (e.g. cyanopropyl or diol) as the stationary phase, with a non-polar mobile phase, is applicable in certain cases.

For analytical separations, the particle size of the most commonly used stationary phases varies between 2 µm and 10 µm. The particles may be spherical or irregular, and of varying porosity and specific surface area. These properties contribute to the chromatographic behaviour of a particular stationary phase. In the case of reversed phases, the nature of the stationary phase, the extent of bonding (e.g. expressed as the carbon loading), and whether the stationary phase is end-capped (i.e. part of the residual silanol groups are silylated) are additional determining factors. Tailing of peaks, particularly of basic substances, can occur when residual silanol groups are present.

In addition to porous particles, superficially porous or monolithic materials may be used.

Unless otherwise prescribed in the monograph, columns made of stainless steel of varying length and internal diameter (Ø) are used for analytical chromatography. Columns with internal diameters of less than 2 mm are often referred to as microbore columns.

The temperature of the mobile phase and the column must be kept constant during the analysis. A column temperature may be specified in the monograph for optimal performance but most separations are performed at 20–25 °C.

MOBILE PHASES

For normal-phase LC, low-polarity organic solvents are generally employed. The residual water content of the solvents used in the mobile phase is to be strictly controlled to obtain reproducible results.

In reversed-phase LC, aqueous mobile phases, usually with organic solvents and/or modifiers, are employed.

The components of the mobile phase are usually filtered to remove particles greater than 0.45 µm in size (or greater than 0.2 µm when the stationary phase is made of sub-2.0 µm particles, and when special detectors, e.g. light-scattering detectors, are used).

Multicomponent mobile phases are prepared by measuring the required volumes (unless masses are specified) of the individual components, followed by mixing. Alternatively, the solvents may be delivered by individual pumps controlled by proportioning valves, by which mixing is performed according to the desired proportion. Solvents are normally degassed before pumping by sparging with helium, sonication and/or using in-line membrane/vacuum modules to avoid the creation of gas bubbles in the detector cell.

Solvents for the preparation of the mobile phase are normally free of stabilisers and, if an ultraviolet detector is employed, are transparent at the wavelength of detection. Solvents and other components employed are to be of appropriate quality. In particular, *water for chromatography R* is used for the preparation of mobile phases when water, or an aqueous solution, is one of the components. Any necessary adjustments of the pH are made to the aqueous component of the mobile phase and not the mixture. If buffer solutions or saline solutions are used, adequate rinsing of the system is carried out with a mixture of water and a small proportion of the organic part of the mobile phase (5 per cent V/V) to prevent crystallisation of salts after completion of the analysis.

Mobile phases may contain other components, for example a counter-ion for ion-pair chromatography or a chiral selector for chiral chromatography using an achiral stationary phase.

DETECTORS

Ultraviolet and visible (UV-Vis) spectrophotometers (including diode array detectors) (2.2.25) are the most commonly employed detectors. Fluorescence spectrophotometers, differential refractometers (RI), electrochemical detectors (ECD), light-scattering detectors, charged aerosol detectors (CAD), mass spectrometers (MS) (2.2.43), radioactivity detectors, multi-angle light-scattering (MALS) detectors or other detectors may be used.

PROCEDURE

Equilibrate the column with the prescribed mobile phase and flow rate, at 20–25 °C or at the temperature specified in the monograph, until a stable baseline is achieved. Prepare the solution(s) of the substance to be examined and the reference solution(s) required. The solutions must be free from solid particles.

Criteria for assessing the suitability of the system are described in general chapter 2.2.46. *Chromatographic separation techniques*. The extent to which adjustments of chromatographic conditions can be made are also given in that general chapter.



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2.2.30. SIZE-EXCLUSION CHROMATOGRAPHY

PRINCIPLE

Size-exclusion chromatography is a liquid chromatography (2.2.29) technique that separates molecules in solution according to their size. With organic mobile phases the technique is known as gel-permeation chromatography and with aqueous mobile phases it is known as gel-filtration chromatography. The sample is introduced into a column, which is filled with a gel or a porous particle packing material, and is carried by the mobile phase through the column. The size separation takes place by repeated exchange of the solute molecules between the solvent of the mobile phase and the same solvent in the stagnant liquid phase (stationary phase) within the pores of the packing material. The pore-size range of the packing material determines the molecular-size range within which separation can occur.

Molecules small enough to penetrate all the pore spaces elute at the total mobile phase volume V_t (also known as total permeation volume). Molecules apparently larger than the maximum pore size of the packing material migrate along the column only through the spaces between the particles of the packing material without being retained and elute at the retention volume of an unretained compound V_0 (also known as exclusion volume or void volume). Separation according to molecular size occurs between the retention volume of an unretained compound and the total mobile phase volume, with useful separation usually occurring in the first two thirds of this range.

EQUIPMENT

The equipment consists essentially of a chromatographic column of varying length and internal diameter ($\emptyset$), if necessary temperature-controlled, packed with a separation material that is capable of fractionation in the appropriate range of molecular sizes. The packing material may be a soft support such as a swollen gel or a rigid support composed of a material such as glass, silica or a solvent-compatible, cross-linked organic polymer. Rigid supports usually require pressurised systems giving faster separations.

One end of the column is usually fitted with a suitable device for applying the sample, such as a flow adapter, a syringe through a septum or an injection valve, and may also be connected to a suitable pump for controlling the flow of the eluent. Alternatively, the sample may be applied directly to the drained bed surface or, where the sample is denser than the eluent, it may be layered beneath the eluent.

The mobile phase is chosen according to sample type, separation medium and technique of detection. The eluent is passed through the column at a constant rate.

The outlet of the column is usually connected to a suitable detector fitted with an automatic recorder that enables the monitoring of the relative concentrations of separated components of the sample. Ultraviolet and visible (UV-Vis) spectrophotometers (2.2.25), differential refractometers (RI), luminescent detectors, multi-angle light-scattering (MALS) detectors and other detectors may be used. An automatic fraction collector may be attached, if necessary.

PROCEDURE

Before carrying out the separation, the packing material is treated and the column packed as described in the monograph, or according to the manufacturer's instructions.

Criteria for assessing the suitability of the system are described in general chapter 2.2.46. *Chromatographic separation techniques*. The extent to which adjustments of chromatographic conditions can be made are also given in that general chapter.

DETERMINATION OF RELATIVE COMPONENT COMPOSITION OF MIXTURES

Carry out the separation as stated in the monograph. If possible, monitor the elution of the components continuously and measure the corresponding peak areas. If the sample is monitored by a physico-chemical property to which all the components of interest exhibit equivalent responses (for example if they have the same specific absorbance), calculate the relative amount of each component by dividing the respective peak area by the sum of the peak areas of all the components of interest. If the responses to the property used for detection of the components of interest are not equivalent, calculate the content by means of calibration curves obtained with the calibration standards prescribed in the monograph.

DETERMINATION OF MOLECULAR MASSES

Size-exclusion chromatography may be used to determine molecular masses by comparison with appropriate calibration standards specified in the monograph. The retention volumes of the calibration standards may be plotted against the logarithm of their molecular masses. The plot usually approximates a straight line within the exclusion and total permeation limits for the separation medium used. From the calibration curve, molecular masses may be estimated. The molecular-mass calibration is valid only for the particular macromolecular solute/solvent system used under the specified experimental conditions.

DETERMINATION OF MOLECULAR SIZE DISTRIBUTION OF POLYMERS

Size-exclusion chromatography may be used to determine the distribution of the molecular size of polymers. However, sample comparison may be valid only for results obtained under the same experimental conditions. The reference standards used for the calibration and the methods for determination of the distribution of molecular sizes of polymers are specified in the monograph.

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2.2.45. SUPERCRITICAL FLUID CHROMATOGRAPHY

PRINCIPLE

Supercritical fluid chromatography (SFC) is a chromatographic separation technique in which the mobile phase is a fluid in a supercritical or a subcritical state. The stationary phase, contained in a column, consists of either finely divided solid particles, such as a silica or porous graphite, a chemically modified stationary phase, as used in liquid chromatography, or, for capillary columns, a cross-linked liquid film evenly coated on the walls of the column.

SFC is based on mechanisms of adsorption or mass distribution.

EQUIPMENT

The apparatus usually consists of a cooled pumping system, an injector, a chromatographic column, contained in an oven, a detector, a pressure regulator and a data acquisition device (or an integrator or a chart recorder).

Pumping system

Pumping systems are required to deliver the mobile phase at a constant flow rate. Pressure fluctuations are to be minimised, e.g. by passing the pressurised solvent through a



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2.2.43. MASS SPECTROMETRY

Mass spectrometry is based on the direct measurement of the ratio of the mass to the number of positive or negative elementary charges of ions (m/z) in the gas phase obtained from the substance to be analysed. This ratio is expressed in atomic mass units (1 a.m.u. = one twelfth the mass of ^{12}C) or in daltons (1 Da = the mass of the hydrogen atom).

The ions, produced in the ion *source* of the apparatus, are accelerated and then separated by the *analyser* before reaching the *detector*. All of these operations take place in a chamber where a pumping system maintains a vacuum of 10^{-3} to 10^{-6} Pa.

The resulting spectrum shows the relative abundance of the various ionic species present as a function of m/z . The signal corresponding to an ion will be represented by several peaks corresponding to the statistical distribution of the various isotopes of that ion. This pattern is called the *isotopic profile* and (at least for small molecules) the peak representing the most abundant isotopes for each atom is called the *monoisotopic peak*.

Information obtained in mass spectrometry is essentially qualitative (determination of the molecular mass, information on the structure from the fragments observed) or quantitative (using internal or external standards) with limits of detection ranging from the picomole to the femtomole.

INTRODUCTION OF THE SAMPLE

The very first step of an analysis is the introduction of the sample into the apparatus without overly disturbing the vacuum. In a common method, called *direct liquid introduction*, the sample is placed on the end of a cylindrical rod (in a quartz crucible, on a filament or on a metal surface). This rod is introduced into the spectrometer after passing through a vacuum lock where a primary intermediate vacuum is maintained between atmospheric pressure and the secondary vacuum of the apparatus.

Other introduction systems allow the components of a mixture to be analysed as they are separated by an appropriate apparatus connected to the mass spectrometer.

Gas chromatography/mass spectrometry. The use of suitable columns (capillary or semi-capillary) allows the end of the column to be introduced directly into the source of the apparatus without using a separator.

Liquid chromatography/mass spectrometry. This combination is particularly useful for the analysis of polar compounds, which are insufficiently volatile or too heat-labile to be analysed by gas chromatography coupled with mass spectrometry. This method is complicated by the difficulty of obtaining ions in the gas phase from a liquid phase, which requires very special interfaces such as:

- *direct liquid introduction*: the mobile phase is nebulised, and the solvent is evaporated in front of the ion source of the apparatus,
- *particle-beam interface*: the mobile phase, which may flow at a rate of up to 0.6 mL/min, is nebulised in a desolvation chamber such that only the analytes, in neutral form, reach the ion source of the apparatus; this technique is used for compounds of relatively low polarity with molecular masses of less than 1000 Da,
- *moving-belt interface*: the mobile phase, which may flow at a rate of up to 1 mL/min, is applied to the surface of a moving belt; after the solvent evaporates, the components

to be analysed are successively carried to the ion source of the apparatus where they are ionised; this technique is rather poorly suited to very polar or heat-labile compounds.

Other types of coupling (electrospray, thermospray, atmospheric-pressure chemical ionisation) are considered to be ionisation techniques in their own right and are described in the section on modes of ionisation.

Supercritical fluid chromatography/mass spectrometry.

The mobile phase, usually consisting of supercritical carbon dioxide enters the gas state after passing a heated restrictor between the column and the ion source.

Capillary electrophoresis/mass spectrometry. The eluent is introduced into the ion source, in some cases after adding another solvent so that flow rates of the order of a few microlitres per minute can be attained. This technique is limited by the small quantities of sample introduced and the need to use volatile buffers.

MODES OF IONISATION

Electron impact. The sample, in the gas state, is ionised by a beam of electrons whose energy (usually 70 eV) is greater than the ionisation energy of the sample. In addition to the molecular ion M^+ , fragments characteristic of the molecular structure are observed. This technique is limited mainly by the need to vaporise the sample. This makes it unsuited to polar, heat-labile or high molecular mass compounds. Electron impact is compatible with the coupling of gas chromatography to mass spectrometry and sometimes with the use of liquid chromatography.

Chemical ionisation. This type of ionisation involves a reagent gas such as methane, ammonia, nitrogen oxide, nitrogen dioxide or oxygen. The spectrum is characterised by ions of the $(M + H)^+$ or $(M - H)^-$ types, or adduct ions formed from the analyte and the gas used. Fewer fragments are produced than with electron impact. A variant of this technique is used when the substance is heat-labile: the sample, applied to a filament, is very rapidly vaporised by the Joule-Thomson effect (desorption chemical ionisation).

Fast-atom bombardment (FAB) or fast-ion bombardment ionisation (liquid secondary-ion mass spectrometry LSIMS). The sample, dissolved in a viscous matrix such as glycerol, is applied to a metal surface and ionised by a beam of neutral atoms such as argon or xenon or high-kinetic-energy caesium ions. Ions of the $(M + H)^+$ or $(M - H)^-$ types or adduct ions formed from the matrix or the sample are produced. This type of ionisation, well suited to polar and heat-labile compounds, allows molecular masses of up to 10 000 Da to be obtained. The technique can be combined with liquid chromatography by adding 1 per cent to 2 per cent of glycerol to the mobile phase; however, the flow rates must be very low (a few microlitres per minute). These ionisation techniques also allow thin-layer chromatography plates to be analysed by applying a thin layer of matrix to the surface of these plates.

Field desorption and field ionisation. The sample is vaporised near a tungsten filament covered with microneedles (*field ionisation*) or applied to this filament (*field desorption*). A voltage of about 10 kV, applied between this filament and a counter-electrode, ionises the sample. These two techniques mainly produce molecular ions M^+ , and $(M + H)^+$ ions and are used for low polarity and/or heat-labile compounds.

Matrix-assisted laser desorption ionisation (MALDI). The sample, in a suitable matrix and deposited on a metal support, is ionised by a pulsed laser beam whose wavelength may range from UV to IR (impulses lasting from a picosecond to a few nanoseconds). This mode of ionisation plays an essential role in the analysis of very high molecular mass compounds (more than 100 000 Da) but is limited to time-of flight analysers (see below).

Electrospray. This mode of ionisation is carried out at atmospheric pressure. The samples, in solution, are introduced into the source through a capillary tube, the end of which has a potential of the order of 5 kV. A gas can be used to facilitate nebulisation. Desolvation of the resulting microdroplets produces singly or multiply charged ions in the gas phase. The flow rates vary from a few microlitres per minute to 1 mL/min. This technique is suited to polar compounds and to the investigation of biomolecules with molecular masses of up to 100 000 Da. It can be coupled to liquid chromatography or capillary electrophoresis.

Atmospheric-pressure chemical ionisation (APCI).

Ionisation is carried out at atmospheric pressure by the action of an electrode maintained at a potential of several kilovolts and placed in the path of the mobile phase, which is nebulised both by thermal effects and by the use of a stream of nitrogen. The resulting ions carry a single charge and are of the $(M + H)^+$ type in the positive mode and of the $(M - H)^-$ type in the negative mode. The high flow rates that can be used with this mode of ionisation (up to 2 mL/min) make this an ideal technique for coupling to liquid chromatography.

Thermospray. The sample, in the mobile phase consisting of water and organic modifiers and containing a volatile electrolyte (generally ammonium acetate) is introduced in nebulised form after having passed through a metal capillary tube at controlled temperature. Acceptable flow rates are of the order of 1 mL/min to 2 mL/min. The ions of the electrolyte ionise the compounds to be analysed. This ionisation process may be replaced or enhanced by an electrical discharge of about 800 volts, notably when the solvents are entirely organic. This technique is compatible with the use of liquid chromatography coupled with mass spectrometry.

ANALYSERS

Differences in the performance of analysers depend mainly on two parameters:

- the range over which m/z ratios can be measured, ie, the *mass range*,
- their *resolving power* characterised by the ability to separate two ions of equal intensity with m/z ratios differing by ΔM , and whose overlap is expressed as a given percentage of valley definition; for example, a resolving power ($M/\Delta M$) of 1000 with 10 per cent valley definition allows the separation of m/z ratios of 1000 and 1001 with the intensity returning to 10 per cent above baseline. However, the resolving power may in some cases (time-of-flight analysers, quadrupoles, ion-trap analysers) be defined as the ratio between the molecular mass and peak width at half height (50 per cent valley definition).

Magnetic and electrostatic analysers. The ions produced in the ion source are accelerated by a voltage V , and focused towards a magnetic analyser (magnetic field B) or an electrostatic analyser (electrostatic field E), depending on the configuration of the instrument. They follow a trajectory of radius r according to Laplace's law:

$$\frac{m}{z} = \frac{B^2 r^2}{2V}$$

Two types of scans can be used to collect and measure the various ions produced by the ion source: a scan of B holding V fixed or a scan of V with constant B . The magnetic analyser is usually followed by an electric sector that acts as a kinetic energy filter and allows the resolving power of the instrument to be increased appreciably. The maximum resolving power of such an instrument (double sector) ranges from 10 000 to 150 000 and in most cases allows the value of m/z ratios to be calculated accurately enough to determine the elemental composition of the corresponding ions. For monocharged ions, the mass range is from 2000 Da to 15 000 Da. Some ions may decompose spontaneously (metastable transitions) or by colliding with a gas (collision-activated dissociation (CAD))

in field-free regions between the ion source and the detector. Examination of these decompositions is very useful for the determination of the structure as well as the characterisation of a specific compound in a mixture and involves tandem mass spectrometry. There are many such techniques depending on the region where these decompositions occur:

- *daughter-ion mode* (determination of the decomposition ions of a given parent ion): $B/E = \text{constant}$, *MIKES* (*Mass-analysed Ion Kinetic Energy Spectroscopy*),
- *parent-ion mode* (determination of all ions which by decomposition give an ion with a specific m/z ratio): $B^2/E = \text{constant}$,
- *neutral-loss mode* (detection of all the ions that lose the same fragment): $B/E(1 - E/E_0)^{1/2} = \text{constant}$, where E_0 is the basic voltage of the electric sector.

Quadrupoles. The analyser consists of four parallel metal rods, which are cylindrical or hyperbolic in cross-section. They are arranged symmetrically with respect to the trajectory of the ions; the pairs diagonally opposed about the axis of symmetry of rods are connected electrically. The potentials to the two pairs of rods are opposed. They are the resultant of a constant component and an alternating component. The ions produced at the ion source are transmitted and separated by varying the voltages applied to the rods so that the ratio of continuous voltage to alternating voltage remains constant. The quadrupoles usually have a mass range of 1 a.m.u. to 2000 a.m.u., but some may range up to 4000 a.m.u. Although they have a lower resolving power than magnetic sector analysers, they nevertheless allow the monoisotopic profile of single charged ions to be obtained for the entire mass range. It is possible to obtain spectra using three quadrupoles arranged in series, Q_1 , Q_2 , Q_3 (Q_2 serves as a collision cell and is not really an analyser; the most commonly used collision gas is argon).

The most common types of scans are the following:

- *daughter-ion mode*: Q_1 selects an m/z ion whose fragments obtained by collision in Q_2 are analysed by Q_3 ,
- *parent-ion mode*: Q_3 filters only a specific m/z ratio, while Q_1 scans a given mass range. Only the ions decomposing to give the ion selected by Q_3 are detected,
- *neutral loss mode*: Q_1 and Q_3 scan a certain mass range but at an offset corresponding to the loss of a fragment characteristic of a product or family of compounds.

It is also possible to obtain spectra by combining quadrupole analysers with magnetic or electrostatic sector instruments; such instruments are called *hybrid mass spectrometers*.

Ion-trap analyser. The principle is the same as for a quadrupole, this time with the electric fields in three dimensions. This type of analyser allows product-ion spectra over several generations (MS^n) to be obtained.

Ion-cyclotron resonance analysers. Ions produced in a cell and subjected to a uniform, intense magnetic field move in circular orbits at frequencies which can be directly correlated to their m/z ratio by applying a Fourier transform algorithm. This phenomenon is called ion-cyclotron resonance. Analysers of this type consist of superconducting magnets and are capable of very high resolving power (up to 1000 000 and more) as well as MS^n spectra. However, very low pressures are required (of the order of 10^{-7} Pa).

Time-of-flight analysers. The ions produced at the ion source are accelerated at a voltage V of 10 kV to 20 kV. They pass through the analyser, consisting of a field-free tube, 25 cm to 1.5 m long, generally called a *flight tube*. The time (t) for an ion to travel to the detector is proportional to the square root of the m/z ratio. Theoretically the mass range of such an analyser is infinite. In practice, it is limited by the ionisation or desorption method. Time-of-flight analysers are mainly used for high molecular mass compounds (up to several hundred thousand daltons). This technique is very sensitive (a

few picomoles of product are sufficient). The accuracy of the measurements and the resolving power of such instruments may be improved considerably by using an electrostatic mirror (reflectron).

SIGNAL ACQUISITION

There are essentially three possible modes.

Complete spectrum mode. The entire signal obtained over a chosen mass range is recorded. The spectrum represents the relative intensity of the different ionic species present as a function of m/z . The results are essentially qualitative. The use of spectral reference libraries for more rapid identification is possible.

Fragmentometric mode (Selected-ion monitoring). The acquired signal is limited to one (single-ion monitoring (SIM)) or several (multiple-ion monitoring (MIM)) ions characteristic of the substance to be analysed. The limit of detection can be considerably reduced in this mode. Quantitative or semiquantitative tests can be carried out using external or internal standards (for example, deuterated standards). Such tests cannot be carried out with time-of-flight analysers.

Fragmentometric double mass spectrometry mode (multiple reaction monitoring (MRM)). The unimolecular or bimolecular decomposition of a chosen precursor ion characteristic of the substance to be analysed is followed specifically. The selectivity and the highly specific nature of this mode of acquisition provide excellent sensitivity levels and make it the most appropriate for quantitative studies using suitable internal standards (for example, deuterated standards). This type of analysis can be performed only on apparatus fitted with three quadrupoles in series, ion-trap analysers or cyclotron-resonance analysers.

CALIBRATION

Calibration allows the corresponding m/z value to be attributed to the detected signal. As a general rule, this is done using a reference substance. This calibration may be external (acquisition file separate from the analysis) or internal (the reference substance(s) are mixed with the substance to be examined and appear on the same acquisition file). The number of ions or points required for reliable calibration depends on the type of analyser and on the desired accuracy of the measurement, for example, in the case of a magnetic analyser where the m/z ratio varies exponentially with the value of the magnetic field, there should be as many points as possible.

SIGNAL DETECTION AND DATA PROCESSING

Ions separated by an analyser are converted into electric signals by a detection system such as a photomultiplier or an electron multiplier. These signals are amplified before being re-converted into digital signals for data processing, allowing various functions such as calibration, reconstruction of spectra, automatic quantification, archiving, creation or use of libraries of mass spectra. The various physical parameters required for the functioning of the apparatus as a whole are controlled by computer.



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2.2.44. TOTAL ORGANIC CARBON IN WATER FOR PHARMACEUTICAL USE

Total organic carbon (TOC) determination is an indirect measure of organic substances present in water for pharmaceutical use. TOC determination can also be used to monitor the performance of various operations in the preparation of medicines.

A variety of acceptable methods is available for determining TOC. Rather than prescribing a given method to be used, this general chapter describes the procedures used to qualify the chosen method and the interpretation of results in limit tests. A standard solution is analysed at suitable intervals, depending on the frequency of measurements; the solution is prepared with a substance that is expected to be easily oxidisable (for example, sucrose) at a concentration adjusted to give an instrument response corresponding to the TOC limit to be measured. The suitability of the system is determined by analysis of a solution prepared with a substance expected to be oxidisable with difficulty (for example, 1,4-benzoquinone).

The various types of apparatus used to measure TOC in water for pharmaceutical use have in common the objective of completely oxidising the organic molecules in the sample water to produce carbon dioxide followed by measurement of the amount of carbon dioxide produced, the result being used to calculate the carbon concentration in the water.

The apparatus used must discriminate between organic and inorganic carbon, the latter being present as carbonate. The discrimination may be effected either by measuring the inorganic carbon and subtracting it from the total carbon, or by purging inorganic carbon from the sample before oxidation. Purging may also entrain organic molecules, but such purgeable organic carbon is present in negligible quantities in water for pharmaceutical use.

Apparatus. Use a calibrated instrument installed either on-line or off-line. Verify the system suitability at suitable intervals as described below. The apparatus must have a limit of detection specified by the manufacturer of 0.05 mg or less of carbon per litre.

TOC water. Use highly purified water complying with the following specifications:

- conductivity: not greater than $1.0 \mu\text{S}\cdot\text{cm}^{-1}$ at 25°C ,
- total organic carbon: not greater than 0.1 mg/L.

Depending on the type of apparatus used, the content of heavy metals and copper may be critical. The manufacturer's instructions should be followed.

Glassware preparation. Use glassware that has been scrupulously cleaned by a method that will remove organic matter. Use TOC water for the final rinse of glassware.

Standard solution. Dissolve sucrose R, dried at 105°C for 3 h in TOC water to obtain a solution containing 1.19 mg of sucrose per litre (0.50 mg of carbon per litre).

Test solution. Using all due care to avoid contamination, collect water to be tested in an airtight container leaving minimal head-space. Examine the water with minimum delay to reduce contamination from the container and its closure.

System suitability solution. Dissolve 1,4-benzoquinone R in TOC water to obtain a solution having a concentration of 0.75 mg of 1,4-benzoquinone per litre (0.50 mg of carbon per litre).

TOC water control. Use TOC water obtained at the same time as that used to prepare the standard solution and the system suitability solution.

Control solutions. In addition to the TOC water control, prepare suitable blank solutions or other solutions needed for establishing the baseline or for calibration adjustments following the manufacturer's instructions; run the appropriate blanks to zero the instrument.

System suitability. Run the following solutions and record the responses: TOC water (r_w); standard solution (r_s); system suitability solution (r_{ss}). Calculate the percentage response efficiency using the expression:

$$\frac{r_{ss} - r_w}{r_s - r_w} \times 100$$

The system is suitable if the response efficiency is not less than 85 per cent and not more than 115 per cent of the theoretical response.

Procedure. Run the test solution and record the response (r_u). The test solution complies with the test if r_u is not greater than $r_s - r_w$.

The method can also be applied using on-line instrumentation that has been adequately calibrated and shown to have acceptable system suitability. The location of instrumentation must be chosen to ensure that the responses are representative of the water used.



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2.2.45. SUPERCRITICAL FLUID CHROMATOGRAPHY

Supercritical fluid chromatography (SFC) is a method of chromatographic separation in which the mobile phase is a fluid in a supercritical or a subcritical state. The stationary phase, contained in a column, consists of either finely divided solid particles, such as a silica or porous graphite, a chemically modified stationary phase, as used in liquid chromatography, or, for capillary columns, a cross-linked liquid film evenly coated on the walls of the column.

SFC is based on mechanisms of adsorption or mass distribution.

APPARATUS

The apparatus usually consists of a cooled pumping system, an injector, a chromatographic column, contained in an oven, a detector, a pressure regulator and a data acquisition device (or an integrator or a chart recorder).

Pumping system

Pumping systems are required to deliver the mobile phase at a constant flow rate. Pressure fluctuations are to be minimised, e.g. by passing the pressurised solvent through a pulse-damping device. Tubing and connections are capable of withstanding the pressures developed by the pumping system.

Microprocessor controlled systems are capable of accurately delivering a mobile phase in either constant or varying conditions, according to a defined programme. In the case of gradient elution, pumping systems which deliver solvent(s) from several reservoirs are available and solvent mixing can be achieved on either the low or high-pressure side of the pump(s).

Injectors

Injection may be carried out directly at the head of the column using a valve.

Stationary phases

Stationary phases are contained in columns which have been described in the chapters on *Liquid chromatography* (2.2.29) (packed columns) and *Gas chromatography* (2.2.28) (capillary columns). A capillary column has a maximum internal diameter ($\varnothing$) of 100 μm .

Mobile phases

Usually the mobile phase is carbon-dioxide which may contain a polar modifier such as methanol, 2-propanol or acetonitrile. The composition, pressure (density), temperature and flow rate of the prescribed mobile phase may either be constant throughout the whole chromatographic procedure (isocratic, isodense, isothermic elution) or may vary according to a defined programme (gradient elution of the modifier, pressure (density), temperature or flow rate).

Detectors

Ultraviolet/visible (UV/Vis) spectrophotometers and flame ionisation detectors are the most commonly employed

detectors. Light scattering detectors, infrared absorption spectrophotometers, thermal conductivity detectors or other special detectors may be used.

METHOD

Prepare the test solution(s) and the reference solution(s) as prescribed. The solutions must be free from solid particles.

Criteria for assessing the suitability of the system are described in the chapter on *Chromatographic separation techniques* (2.2.46). The extent to which adjustments of parameters of the chromatographic system can be made to satisfy the criteria of system suitability are also given in this chapter.



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2.2.46. CHROMATOGRAPHIC SEPARATION TECHNIQUES⁽⁷⁾

INTRODUCTION

Chromatographic separation techniques are multi-stage separation methods in which the components of a sample are distributed between 2 phases, one of which is stationary, while the other is mobile. The stationary phase may be a solid or a liquid supported on a solid or a gel. The stationary phase may be packed in a column, spread as a layer, or distributed as a film, etc. The mobile phase may be gaseous or liquid. The separation may be based on adsorption, mass distribution (partition), ion exchange, etc., or may be based on differences in the physico-chemical properties of the molecules such as size, mass, volume, etc.

This chapter contains definitions and calculations of common parameters and generally applicable requirements for system suitability. Principles of separation, equipment and methods are given in the corresponding general chapters:

- ◇ – paper chromatography (2.2.26);
- thin-layer chromatography (2.2.27);
- gas chromatography (2.2.28);
- liquid chromatography (2.2.29);
- size-exclusion chromatography (2.2.30).◇

DEFINITIONS

The system suitability and acceptance criteria in monographs have been set using parameters as defined below. With some equipment, certain parameters such as signal-to-noise ratio and resolution can be calculated using software provided by the manufacturer. It is the responsibility of the user to ensure that the calculation methods used in the software are equivalent to the requirements of the European Pharmacopoeia and to make any necessary corrections if this is not the case.

Chromatogram

A graphical or other representation of detector response, effluent concentration or other quantity used as a measure of effluent concentration, versus time or volume. Ideally, chromatograms are represented as a sequence of Gaussian peaks on a baseline (Figure 2.2.46.-1).

- V_M = hold-up volume;
- t_M = hold-up time;
- V_{R1} = retention volume of peak 1;
- t_{R1} = retention time of peak 1;
- V_{R2} = retention volume of peak 2;
- t_{R2} = retention time of peak 2;

⁽⁷⁾ This chapter has undergone pharmacopoeial harmonisation. See chapter 5.8. *Pharmacopoeial harmonisation*.

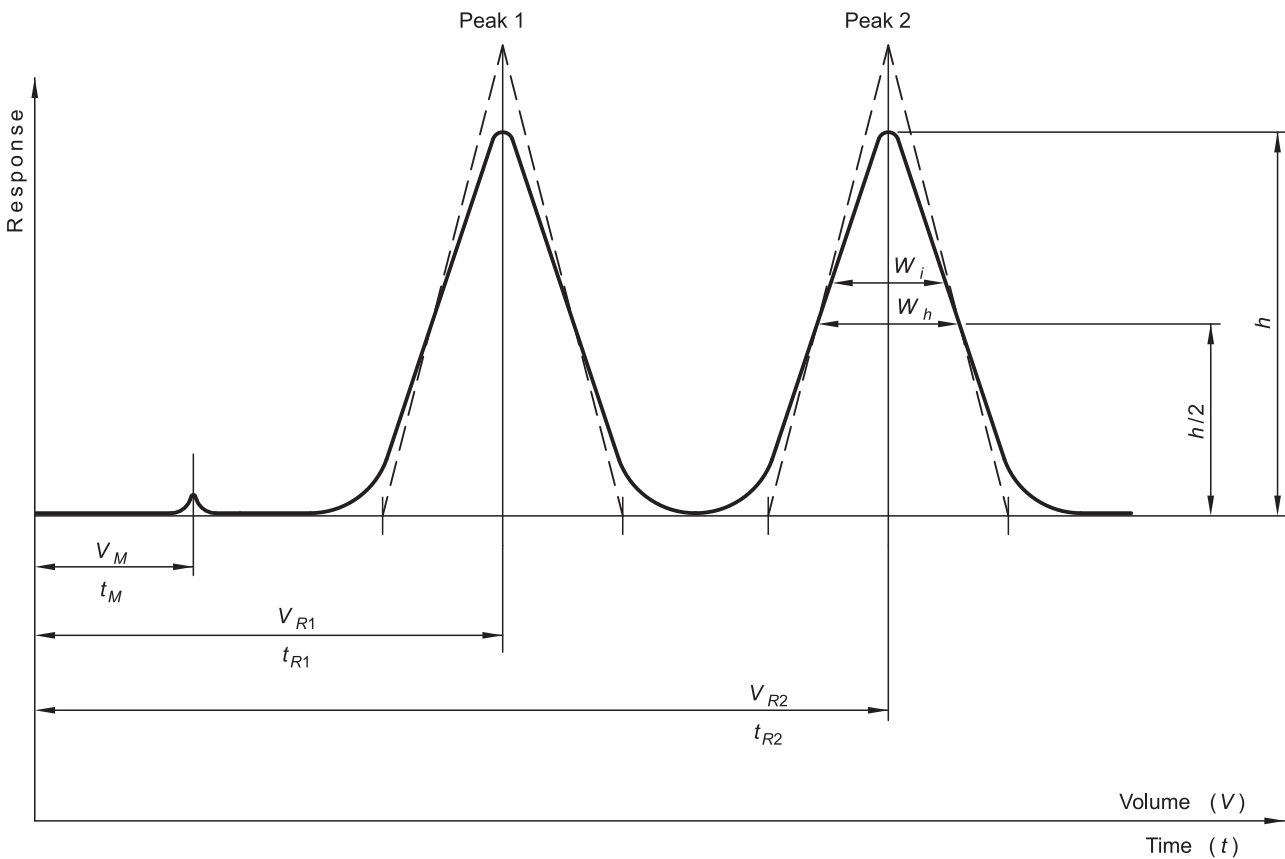


Figure 2.2.46.-1.

W_h = peak width at half-height;
 W_i = peak width at the inflexion point;
 h = height of the peak;
 $h/2$ = half-height of the peak.

Distribution constant (K_0)

In size-exclusion chromatography, the elution characteristics of a component in a particular column may be given by the distribution constant (also referred to as distribution coefficient), which is calculated using the following equation:

$$K_0 = \frac{t_R - t_0}{t_t - t_0}$$

t_R = retention time;
 t_0 = retention time of an unretained compound;
 t_t = total mobile phase time.

Dwell volume (D) (also referred to as V_D)

The dwell volume (also known as gradient delay volume) is the volume between the point at which the eluents meet and the inlet of the column. It can be determined using the following procedure.

Column: replace the chromatographic column by an appropriate capillary tubing (e.g. 1 m $\times$ 0.12 mm).

Mobile phase:

- mobile phase A: water for chromatography R;
- mobile phase B: 0.1 per cent V/V solution of acetone R in water for chromatography R;

Time (min)	Mobile phase A (per cent V/V)	Mobile phase B (per cent V/V)
0 - 20	100 $\rightarrow$ 0	0 $\rightarrow$ 100
20 - 30	0	100

Flow rate: set to obtain sufficient back-pressure (e.g. 2 mL/min).

Detection: spectrophotometer at 265 nm.

Determine the time ($t_{0.5}$), in minutes, when the absorbance has increased by 50 per cent (Figure 2.2.46.-2).

$$D = t_D \times F$$

t_D = $t_{0.5} - 0.5t_G$ in minutes;
 t_G = pre-defined gradient time (= 20 min);
 F = flow rate, in millilitres per minute.

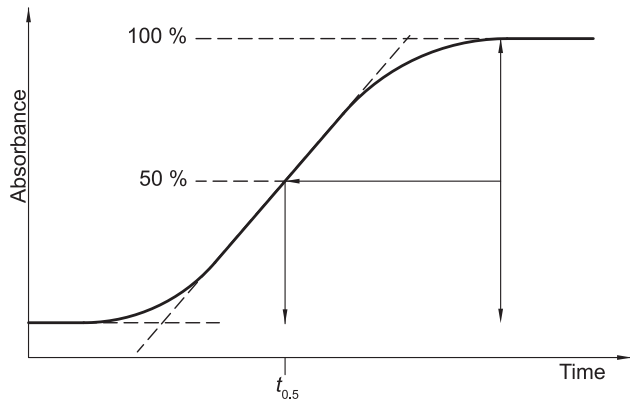


Figure 2.2.46.-2.

Note: where applicable, this measurement is performed with the autosampler in the 'inject' position so as to include the injection loop volume in the dwell volume.

Hold-up time (t_M)

Time required for elution of an unretained component (Figure 2.2.46.-1, baseline scale being in minutes or seconds).

In size-exclusion chromatography, the term 'retention time of an unretained compound (t_0)' is used.

Hold-up volume (V_M)

Volume of the mobile phase required for elution of an unretained component. It may be calculated from the hold-up time (t_M) and the flow rate (F) in millilitres per minute using the following equation:

$$V_M = t_M \times F$$

In size-exclusion chromatography, the term 'retention volume of an unretained compound (V_0)' is used.

Peak

Portion of a chromatogram recording the detector response when a single component (or 2 or more unresolved components) is eluted from the column.

The peak response may be represented by the peak area or the peak height (h).

Peak-to-valley ratio (p/v)

The peak-to-valley ratio may be employed as a system suitability criterion when baseline separation between 2 peaks is not achieved (Figure 2.2.46.-3).

$$p/v = \frac{H_p}{H_v}$$

H_p = height above the extrapolated baseline of the minor peak;

H_v = height above the extrapolated baseline at the lowest point of the curve separating the minor and major peaks.

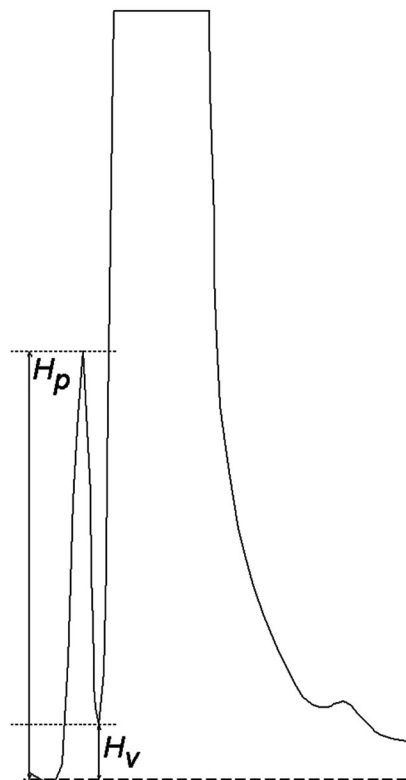


Figure 2.2.46.-3.

Plate height (H) (synonym: Height equivalent to one theoretical plate (HETP))

Ratio of the column length (L), in micrometres, to the plate number (N).

$$H = \frac{L}{N}$$

Plate number (N) (synonym: Number of theoretical plates)

A number indicative of column performance (column efficiency). It can only be calculated from data obtained under either isothermal, isocratic or isodense conditions, depending on the technique, as the plate number, using the following equation, the values of t_R and w_h being expressed in the same units:

$$N = 5.54 \left(\frac{t_R}{w_h} \right)^2$$

t_R = retention time of the peak corresponding to the component;

w_h = peak width at half-height ($h/2$).

The plate number varies with the component as well as with the column, the column temperature, the mobile phase and the retention time.

Reduced plate height (h)

Ratio of the plate height (H), in micrometres, to the particle diameter (d_p) in micrometres.

$$h = \frac{H}{d_p}$$

Relative retardation (R_{rel})

The relative retardation, used in planar chromatography, is calculated as the ratio of the distances travelled by the spot of the component of interest and a reference component (Figure 2.2.46.-4).

$$R_{rel} = \frac{b}{c}$$

a = migration distance of the mobile phase;

b = migration distance of the component of interest;

c = migration distance of the reference component.

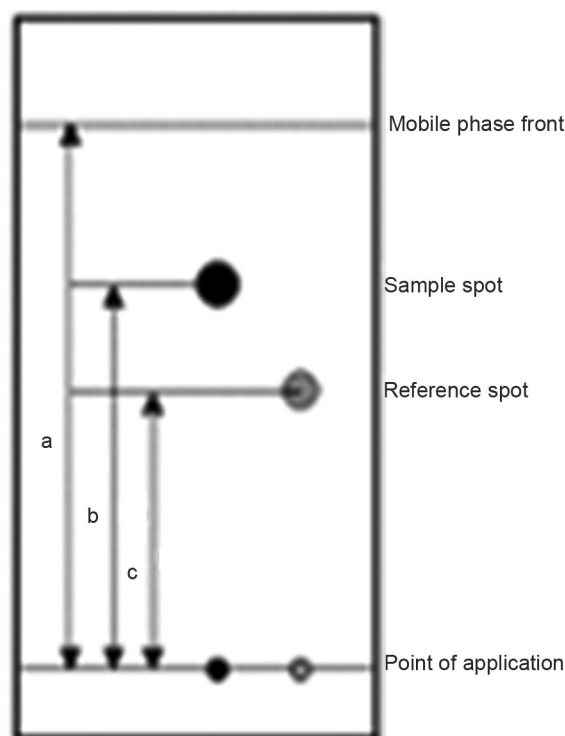


Figure 2.2.46.-4.

Relative retention (*r*)

Relative retention is calculated as an estimate using the following equation:

$$r = \frac{t_{Ri} - t_M}{t_{Rst} - t_M}$$

- t_{Ri} = retention time of the peak of interest;
 t_{Rst} = retention time of the reference peak (usually the peak corresponding to the substance to be examined);
 t_M = hold-up time.

Relative retention, unadjusted (*r_G*)

Unadjusted relative retention is calculated using the following equation:

$$r_G = \frac{t_{Ri}}{t_{Rst}}$$

Unless otherwise indicated, values for relative retention stated in monographs correspond to unadjusted relative retention.

◇ In planar chromatography, the retardation factors R_{Fst} and R_{Fi} are used instead of t_{Rst} and t_{Ri} . R_{Fst} (also known as R_{st}) indicates the ratio of the distance travelled by a substance to the distance travelled by a substance used as reference. ◇

Resolution (*R_s*)

The resolution between peaks of 2 components (Figure 2.2.46.-1) may be calculated using the following equation:

$$R_s = \frac{1.18 (t_{R2} - t_{R1})}{w_{h1} + w_{h2}}$$

- $t_{R2} > t_{R1}$
 t_{R1}, t_{R2} = retention times of the peaks;
 w_{h1}, w_{h2} = peak widths at half-height.

In quantitative planar chromatography, using densitometry, the migration distances are used instead of retention times and the resolution between peaks of 2 components may be calculated using the following equation:

$$R_s = \frac{1.18a (R_{F2} - R_{F1})}{w_{h1} + w_{h2}}$$

- $R_{F2} > R_{F1}$
 R_{F1}, R_{F2} = retardation factors of the peaks;
 w_{h1}, w_{h2} = peak widths at half-height;
 a = migration distance of the solvent front.

Retardation factor (*R_F*)

The retardation factor ◇ (also known as retention factor (R_f)) ◇, used in planar chromatography, is the ratio of the distance from the point of application to the centre of the spot and the distance simultaneously travelled by the solvent front from the point of application (Figure 2.2.46.-4).

$$R_F = \frac{b}{a}$$

- b = migration distance of the component of interest;
 a = migration distance of the solvent front.

Retention factor (*k*)

The retention factor (also known as mass distribution ratio (D_m) or capacity factor (k')) is defined as:

$$k = \frac{\text{amount of component in stationary phase}}{\text{amount of component in mobile phase}} = K_C \times \frac{V_S}{V_M}$$

- K_C = distribution constant (also known as equilibrium distribution coefficient);
 V_S = volume of the stationary phase;
 V_M = volume of the mobile phase.

The retention factor of a component may be determined from the chromatogram using the following equation:

$$k = \frac{t_R - t_M}{t_M}$$

- t_R = retention time;
 t_M = hold-up time.

Retention time (*t_R*)

Time elapsed between the injection of the sample and the appearance of the maximum peak response of the eluted sample zone (Figure 2.2.46.-1, baseline scale being in minutes or seconds).

Retention volume (*V_R*)

Volume of the mobile phase required for elution of a component. It may be calculated from the retention time (t_R) and the flow rate (F) in millilitres per minute using the following equation:

$$V_R = t_R \times F$$

Retention time of an unretained compound (*t₀*)

In size-exclusion chromatography, retention time of a component whose molecules are larger than the largest gel pores (Figure 2.2.46.-5).

Retention volume of an unretained compound (*V₀*)

In size-exclusion chromatography, retention volume of a component whose molecules are larger than the largest gel pores. It may be calculated from the retention time of an unretained compound (t_0) and the flow rate (F) in millilitres per minute using the following equation:

$$V_0 = t_0 \times F$$

Separation factor (*α*)

Relative retention calculated for 2 adjacent peaks (by convention, the value of the separation factor is always > 1) using the following equation:

$$\alpha = \frac{k_2}{k_1}$$

- k_1 = retention factor of peak 1;
 k_2 = retention factor of peak 2.

Signal-to-noise ratio (*S/N*)

Short-term noise influences the precision and accuracy of quantitation. The signal-to-noise ratio is calculated using the following equation:

$$S/N = \frac{2H}{h}$$

- H = height of the peak (Figure 2.2.46.-6) corresponding to the component concerned, in the chromatogram obtained with the reference solution, measured from the maximum of the peak to the extrapolated baseline of the signal observed over a distance equal to 20 times the width at half-height;
 h = range of the noise in a chromatogram obtained after injection of a blank (Figure 2.2.46.-7), observed over a distance equal to 20 times the width at half-height of the peak in the chromatogram obtained with the reference solution and, if possible, situated equally around the place where this peak would be found.

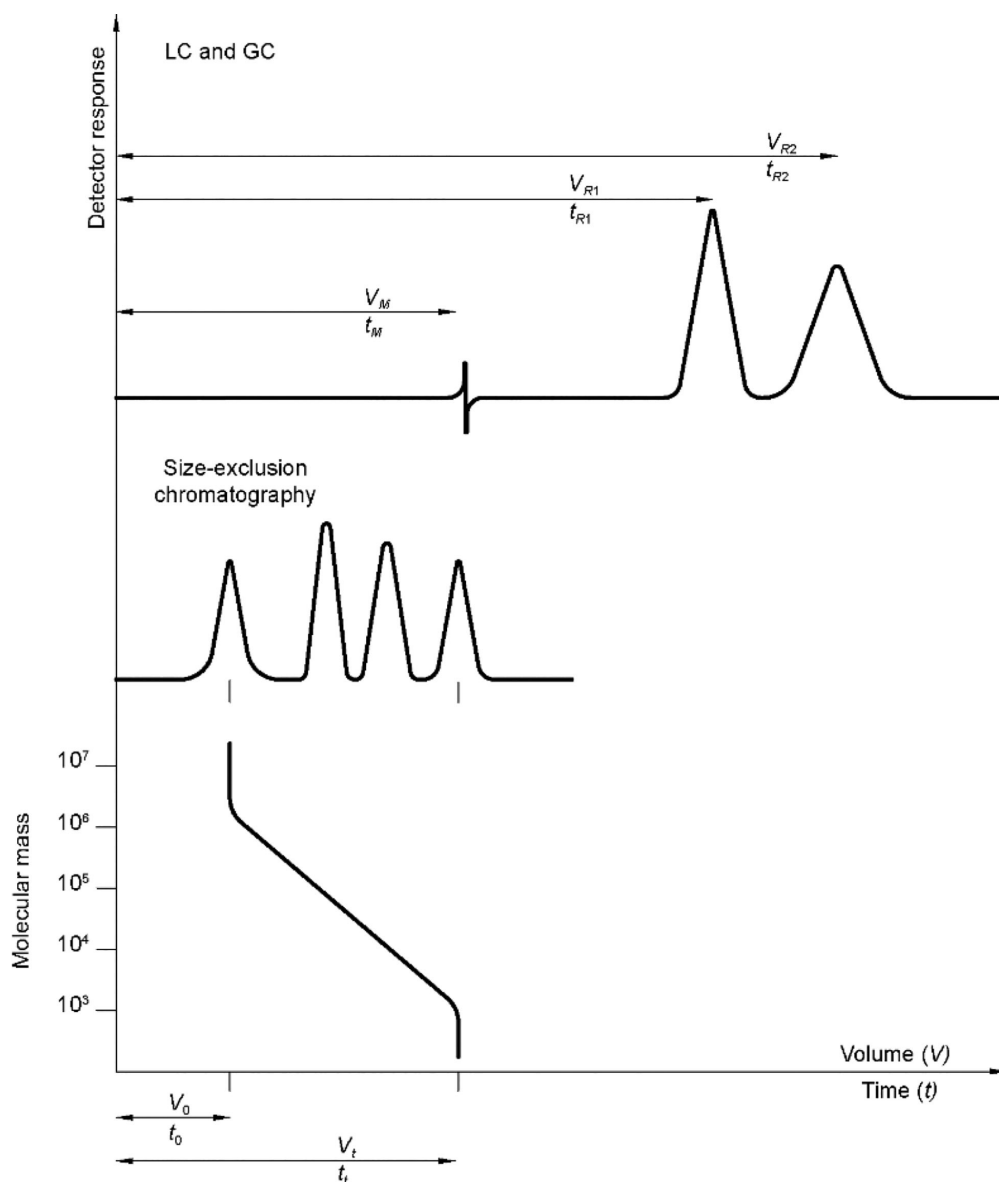


Figure 2.2.46-5.

If a baseline of 20 times the width at half-height is not obtainable because of peaks due to solvents or reagents, or arising from the mobile phase or the sample matrix, or due to the gas chromatography temperature programme, a baseline of at least 5 times the width at half-height is permitted.

Symmetry factor (A_s)

The symmetry factor of a peak (also known as asymmetry factor or tailing factor) (Figure 2.2.46-8) is calculated using the following equation:

$$A_s = \frac{w_{0.05}}{2d}$$

$w_{0.05}$ = width of the peak at one-twentieth of the peak height;

d = distance between the perpendicular dropped from the peak maximum and the leading edge of the peak at one-twentieth of the peak height.

An A_s value of 1.0 signifies symmetry. When $A_s > 1.0$, the peak is tailing. When $A_s < 1.0$, the peak is fronting.

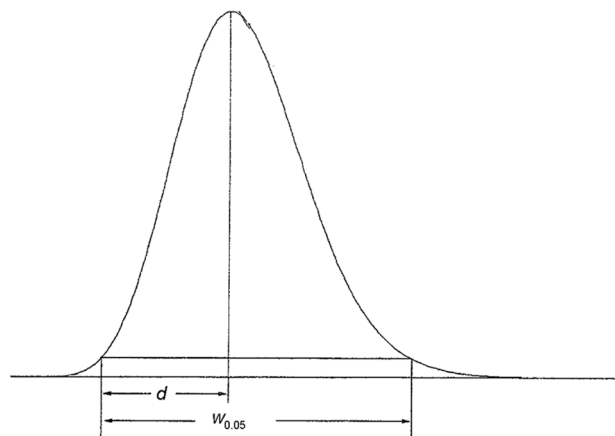


Figure 2.2.46-8.

System repeatability

The repeatability of response is expressed as an estimated percentage relative standard deviation (%RSD) of a consecutive series of measurements for not fewer than 3 injections or applications of a reference solution, and is calculated using the following equation:

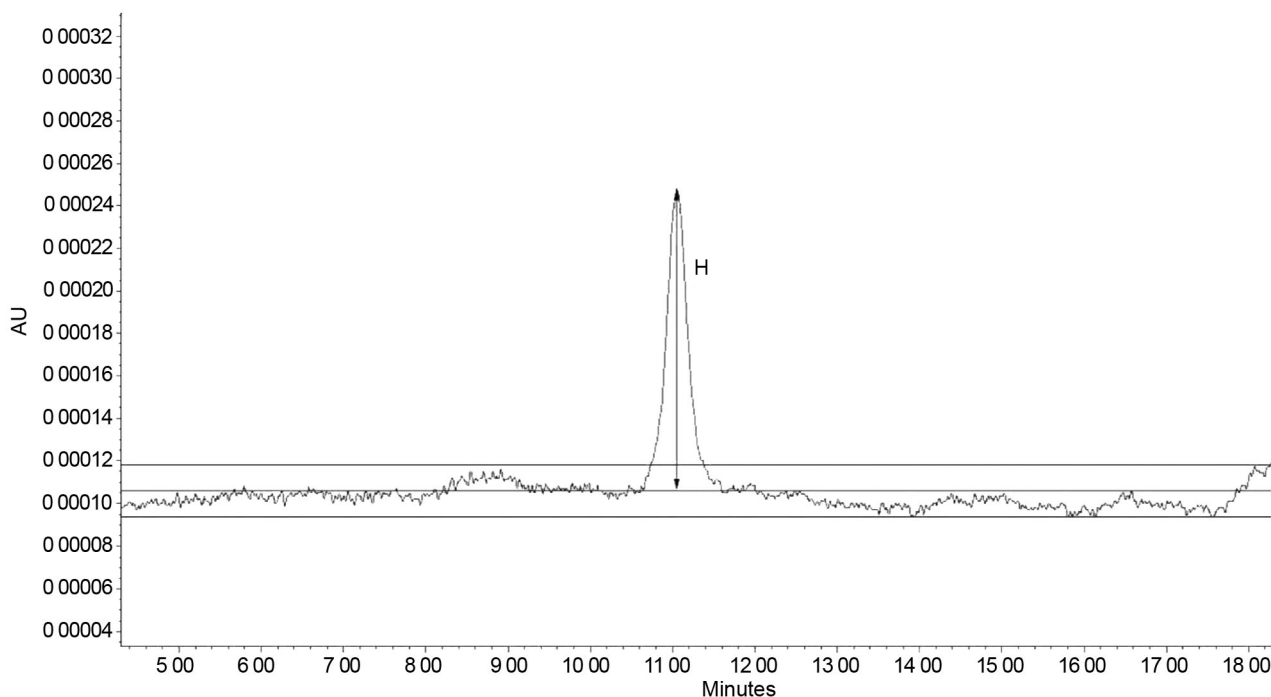


Figure 2.2.46.-6. – Chromatogram of the reference solution

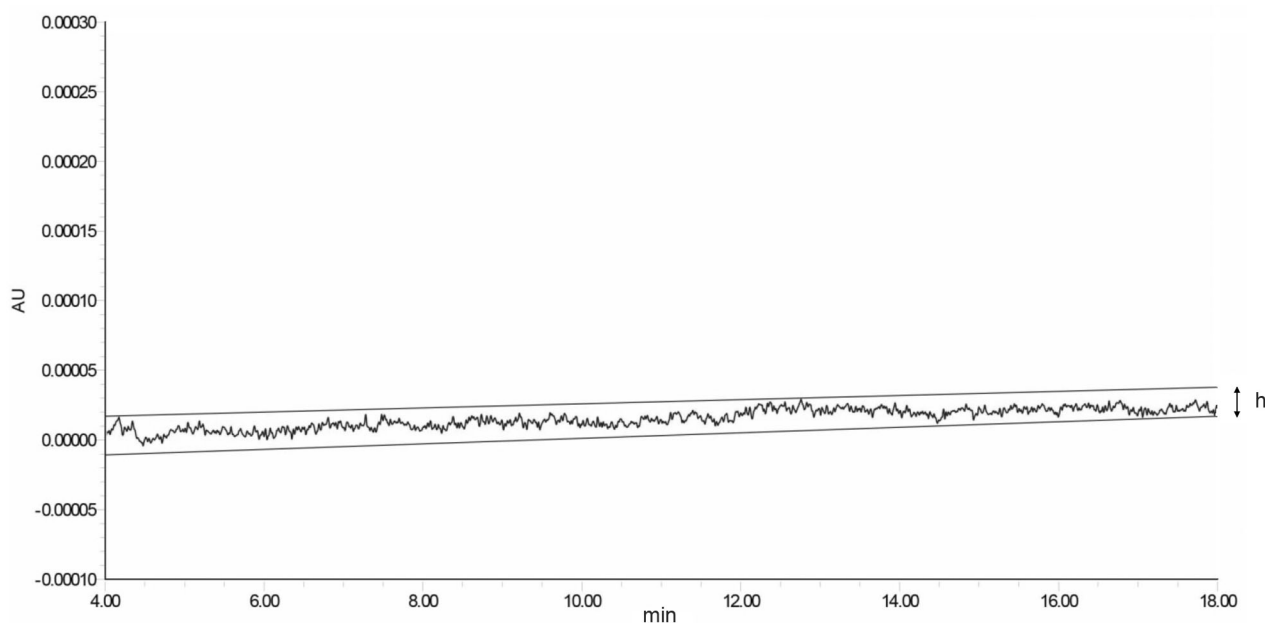


Figure 2.2.46.-7. – Chromatogram of a blank

$$\% RSD = \frac{100}{\bar{y}} \sqrt{\frac{\sum (y_i - \bar{y})^2}{n - 1}}$$

- y_i = individual values expressed as peak area, peak height, or ratio of areas by the internal standardisation method;
 $\bar{y}$ = mean of individual values;
 n = number of individual values.

Total mobile phase time (t_t)

In size-exclusion chromatography, retention time of a component whose molecules are smaller than the smallest gel pores (Figure 2.2.46.-5).

Total mobile phase volume (V_t)

In size-exclusion chromatography, retention volume of a component whose molecules are smaller than the smallest

gel pores. It may be calculated from the total mobile phase time (t_t) and the flow rate (F) in millilitres per minute using the following equation:

$$V_t = t_t \times F$$

SYSTEM SUITABILITY

This section only covers liquid chromatography and gas chromatography.

The various components of the equipment employed must be qualified and be capable of achieving the performance required to conduct the test or assay.

The system suitability tests represent an integral part of the analytical procedure and are used to ensure adequate performance of the chromatographic system. Column plate number, retention factor (mass distribution ratio), system repeatability, signal-to-noise, resolution/peak-to-valley ratio and symmetry factor are the parameters that may be employed

in assessing the performance of the chromatographic system. When stated in the individual monograph, in case of complex chromatographic profiles (e.g. for biotechnological/biological products), visual comparison of the profiles can be used as a system suitability test.

Factors that may affect the chromatographic behaviour include:

- composition and temperature of the mobile phase;
- ionic strength and pH of the aqueous component of the mobile phase;
- flow rate, column dimensions, column temperature and pressure;
- stationary phase characteristics including type of chromatographic support (particle-based or monolithic), particle or pore size, porosity, specific surface area;
- reversed phase and other surface-modification of the stationary phases, extent of chemical modification (as expressed by end-capping, carbon loading, etc.).

When retention times and relative retentions are provided in monographs, they are for information purposes only, unless otherwise stated in the monograph. There are no acceptance criteria applied to relative retentions.

Compliance with the system suitability criteria is required throughout the chromatographic procedure. No sample analysis is acceptable unless the suitability of the system has been demonstrated.

The following requirements are to be fulfilled, in addition to any other system suitability criteria stated in the monograph. When specific requirements are stated in the monograph, they supersede the requirements mentioned in this chapter.

System repeatability – assay of an active substance or an excipient

In an assay of an active substance or an excipient, where the target value is 100 per cent for a pure substance and no system repeatability requirement is specified, the maximum permitted relative standard deviation (%RSD_{max}) for the defined limits is calculated for a series of injections of the reference solution ($n = 3-6$). The maximum permitted relative standard deviation of the peak response does not exceed the appropriate value given in Table 2.2.46.-1.

$$\%RSD_{max} = \frac{KB\sqrt{n}}{t_{90\%,n-1}}$$

- K = constant (0.349), obtained from the expression $K = \frac{0.6}{\sqrt{2}} \times \frac{t_{90\%,5}}{\sqrt{6}}$ in which $\frac{0.6}{\sqrt{2}}$ represents the required percentage relative standard deviation determined on 6 injections for $B = 1.0$;
- B = upper limit given in the definition of the individual monograph minus 100 per cent;
- n = number of replicate injections of the reference solution ($3 \leq n \leq 6$);
- $t_{90\%,n-1}$ = Student's t at the 90 per cent probability level (double sided) with $n-1$ degrees of freedom.

Table 2.2.46.-1. – Maximum permitted relative standard deviation (assay)

	Number of individual injections n			
	3	4	5	6
B (per cent)	Maximum permitted relative standard deviation (per cent)			
2.0	0.41	0.59	0.73	0.85
2.5	0.52	0.74	0.92	1.06
3.0	0.62	0.89	1.10	1.27

System sensitivity

The signal-to-noise ratio is used to define the system sensitivity. The limit of quantitation (corresponding to a signal-to-noise ratio of 10) is equal to or less than the reporting threshold.

◊ To determine the signal-to-noise ratio, inject a solution of the substance to be examined at the concentration corresponding to the reporting threshold (e.g. 0.05 per cent). Alternatively, use the reference solution used for the quantitation of unspecified impurities (e.g. 0.10 per cent of the concentration of the test solution) and extrapolate the area of the principal peak to the reporting threshold (e.g. half the peak area obtained with the reference solution at 0.10 per cent). ◊

Peak symmetry

Unless otherwise stated, in a test or assay, the symmetry factor (tailing factor) of the peak used for quantitation is 0.8-1.8.

ADJUSTMENT OF CHROMATOGRAPHIC CONDITIONS

The chromatographic conditions described have been validated during the elaboration of the monograph.

The extent to which the various parameters of a chromatographic test may be adjusted without fundamentally modifying the pharmacopoeial analytical procedures are listed below. Changes other than those indicated require revalidation of the procedure.

Multiple adjustments can have a cumulative effect on the performance of the system and are to be properly evaluated by the users. This is particularly important in cases where the separation pattern is described as a profile. In those cases, a risk assessment has to be carried out.

Any adjustments must be made on the basis of the pharmacopoeial procedure.

If adjustments are made to a pharmacopoeial procedure, additional verification tests may be required. To verify the suitability of the adjusted pharmacopoeial procedure, assess the relevant analytical performance characteristics potentially affected by the change.

When a pharmacopoeial analytical procedure has been adjusted according to the requirements stated below, no further adjustments are allowed without appropriate revalidation.

Compliance with the system suitability criteria is required to verify that conditions for satisfactory performance of the test or assay are achieved.

Adjustment of conditions with gradient elution (HPLC) or temperature programming (GC) is more critical than with isocratic (HPLC) or isothermal (GC) elution, since it may shift some peaks to a different step of the gradient or to different elution temperatures, potentially causing partial or complete coelution of adjacent peaks or peak inversion, and thereby leading to the incorrect assignment of peaks and the masking of peaks, or a shift such that elution occurs beyond the prescribed elution time.

For some parameters, the adjustments are explicitly defined in the monograph to ensure the system suitability.

THIN-LAYER CHROMATOGRAPHY

Composition of the mobile phase: the amount of the minor solvent components may be adjusted by ± 30 per cent relative or ± 2 per cent absolute, whichever is the larger; no other component is altered by more than 10 per cent absolute. A minor component comprises less than or equal to $(100/n)$ per cent, n being the total number of components of the mobile phase. For a minor component at 10 per cent of the mobile phase, a 30 per cent relative adjustment allows a range of 7-13 per cent whereas a 2 per cent absolute adjustment allows a range of 8-12 per cent, the relative value therefore being the larger; for a minor component at 5 per cent of the mobile phase, a 30 per cent relative adjustment allows a range of

3.5-6.5 per cent whereas a 2 per cent absolute adjustment allows a range of 3-7 per cent, the absolute value being the larger in this case.

pH of the aqueous component of the mobile phase: ± 0.2 pH units, unless otherwise prescribed.

Concentration of salts in the buffer component of a mobile phase: ± 10 per cent.

Application volume: 10-20 per cent of the prescribed volume if using fine particle size plates (2-10 μm).

LIQUID CHROMATOGRAPHY: ISOCRATIC ELUTION

Column parameters and flow rate

Stationary phase: no change of the identity of the substituent (e.g. no replacement of C18 by C8); the other physico-chemical characteristics of the stationary phase (i.e. chromatographic support, surface modification and extent of chemical modification) must be similar; a change from totally porous particle (TPP) columns to superficially porous particle (SPP) columns is allowed provided the above-mentioned requirements are met.

Column dimensions (particle size, length): the particle size and/or length of the column may be modified provided that the ratio of the column length (L) to the particle size (dp) remains constant or in the range -25 per cent to $+50$ per cent of the prescribed L/dp ratio. For the application of particle-size adjustment from totally porous to superficially porous particles, other combinations of L and dp can be used provided that the plate number (N) is within -25 per cent to $+50$ per cent relative to the prescribed column.

These changes are acceptable provided the system suitability requirements are fulfilled and the selectivity and elution order of the specified impurities to be controlled are demonstrated to be equivalent.

Column dimensions (internal diameter): in the absence of a change in particle size and/or length of the column, the internal diameter of the column may be adjusted.

Caution is necessary when the adjustment results in smaller peak volumes, due to a smaller particle size or a smaller internal diameter, a situation that may require adjustments to minimise extra-column band broadening by factors such as instrument connections, detector cell volume and sampling rate, and injection volume.

When the particle size is changed, the flow rate requires adjustment, because smaller-particle columns will require higher linear velocities for the same performance (as measured by reduced plate height). Flow rate is adjusted for changes in column diameter and particle size using the following equation:

$$F_2 = F_1 \times \frac{dc_2^2 \times dp_1}{dc_1^2 \times dp_2}$$

- F_1 = flow rate indicated in the monograph, in millilitres per minute;
- F_2 = adjusted flow rate, in millilitres per minute;
- dc_1 = internal diameter of the column indicated in the monograph, in millimetres;
- dc_2 = internal diameter of the column used, in millimetres;
- dp_1 = particle size indicated in the monograph, in micrometres;
- dp_2 = particle size of the column used, in micrometres.

When a change is made from $\geq 3 \mu\text{m}$ to $< 3 \mu\text{m}$ particles in isocratic elution, an additional increase in linear velocity (by adjusting the flow rate) may be justified, provided that the column performance does not drop by more than 20 per cent. Similarly, when a change is made from $< 3 \mu\text{m}$ to $\geq 3 \mu\text{m}$

particles, an additional reduction of linear velocity (flow rate) may be justified to avoid reduction in column performance by more than 20 per cent.

After an adjustment due to a change in column dimensions, an additional change in flow rate of ± 50 per cent is permitted.

Column temperature: $\pm 10^\circ\text{C}$, where the operating temperature is specified, unless otherwise prescribed.

Further adjustments in analytical procedure conditions (mobile phase, temperature, pH, etc.) may be required, within the permitted ranges described under System Suitability and Adjustment of chromatographic conditions in this chapter.

Mobile phase

Composition: the amount of the minor solvent components may be adjusted by ± 30 per cent relative (see example under Thin-layer chromatography); no component is altered by more than 10 per cent absolute. A minor component comprises less than $(100/n)$ per cent, n being the total number of components of the mobile phase.

pH of the aqueous component of the mobile phase: ± 0.2 pH units, unless otherwise prescribed.

Concentration of salts in the buffer component of a mobile phase: ± 10 per cent.

Flow rate: in the absence of a change in column dimensions, an adjustment of the flow rate by ± 50 per cent is permitted.

Detector wavelength: no adjustment permitted.

Injection volume: when the column dimensions are changed, the following equation may be used for adjusting the injection volume:

$$V_{inj2} = V_{inj1} \times \frac{L_2 \times dc_1^2}{L_1 \times dc_2^2}$$

- V_{inj1} = injection volume indicated in the monograph, in microlitres;
- V_{inj2} = adjusted injection volume, in microlitres;
- L_1 = column length indicated in the monograph, in millimetres;
- L_2 = new column length, in millimetres;
- dc_1 = column internal diameter indicated in the monograph, in millimetres;
- dc_2 = new column internal diameter, in millimetres.

This equation may not be applicable to changes from TPP columns to SPP columns.

Even in the absence of any column dimension change, the injection volume may be varied provided the system suitability criteria remain within their established acceptability limits. When the injection volume is decreased, special attention is given to (limit of) detection and repeatability of the peak response(s) to be determined. An increase is permitted provided that, in particular, linearity and resolution of the peak(s) to be determined remain satisfactory.

LIQUID CHROMATOGRAPHY: GRADIENT ELUTION

Adjustment of chromatographic conditions for gradient systems requires greater caution than for isocratic systems.

Column parameters and flow rate

Stationary phase: no change of the identity of the substituent (e.g. no replacement of C18 by C8); the other physico-chemical characteristics of the stationary phase (i.e. chromatographic support, surface modification and extent of chemical modification) must be similar; a change from totally porous particle (TPP) columns to superficially porous particle (SPP) columns is allowed provided the above-mentioned requirements are met.

Column dimensions (particle size, length): the particle size and/or length of the column may be modified provided that the ratio of the column length (L) to the particle size (dp) remains constant or in the range -25 per cent to $+50$ per

cent of the prescribed L/dp ratio. For the application of particle-size adjustment from totally porous to superficially porous particles, other combinations of L and dp can be used provided that the ratio $(t_R/w_h)^2$ is within – 25 per cent to + 50 per cent relative to the prescribed column, for each peak used to check the system suitability, as stated in this chapter and the individual monograph.

These changes are acceptable provided the system suitability requirements are fulfilled and the selectivity and elution order of the specified impurities to be controlled are demonstrated to be equivalent.

Column dimensions (internal diameter): in the absence of a change in particle size and/or length of the column, the internal diameter of the column may be adjusted.

Caution is necessary when the adjustment results in smaller peak volumes, due to a smaller particle size or a smaller internal diameter, a situation that may require adjustments to minimise extra-column band broadening by factors such as instrument connections, detector cell volume and sampling rate, and injection volume.

When the particle size is changed, the flow rate requires adjustment, because smaller-particle columns will require higher linear velocities for the same performance (as measured by reduced plate height). Flow rate is adjusted for changes in column diameter and particle size using the following equation:

$$F_2 = F_1 \times \frac{dc_2^2 \times dp_1}{dc_1^2 \times dp_2}$$

- F_1 = flow rate indicated in the monograph, in millilitres per minute;
- F_2 = adjusted flow rate, in millilitres per minute;
- dc_1 = internal diameter of the column indicated in the monograph, in millimetres;
- dc_2 = internal diameter of the column used, in millimetres;
- dp_1 = particle size indicated in the monograph, in micrometres;
- dp_2 = particle size of the column used, in micrometres.

A change in column dimensions, and thus in column volume, impacts the gradient volume, which controls selectivity. Gradients are adjusted to the column volume by changing the gradient volume in proportion to the column volume. This applies to every gradient segment volume. Since the gradient volume is the gradient time (t_G) multiplied by the flow rate (F), the gradient time for each gradient segment must be adjusted to maintain a constant ratio of the gradient volume to the column volume (expressed as $L \times dc^2$). Thus, the new gradient time (t_{G2}) can be calculated from the original gradient time (t_{G1}), the flow rate(s) and the column dimensions as follows:

$$t_{G2} = t_{G1} \times \frac{F_1}{F_2} \times \frac{L_2 \times dc_2^2}{L_1 \times dc_1^2}$$

Thus, the change in conditions for gradient elution requires 3 steps:

- (1) adjust the column length and particle size according to L/dp ;
- (2) adjust the flow rate for changes in particle size and column diameter; and
- (3) adjust the gradient time of each segment for changes in column length, diameter and flow rate.

The example below illustrates this process.

Table 2.2.46.-2. – Example of adjustments for liquid chromatography - gradient elution

Variable	Original conditions	Adjusted conditions	Comment
Column length (L) in mm	150	100	User's choice
Column diameter (dc) in mm	4.6	2.1	User's choice
Particle size (dp) in μm	5	3	User's choice
L/dp	30.0	33.3	(1)
Flow rate (F) in mL/min	2.0	0.7	(2)
Gradient adjustment factor (t_{G2}/t_{G1})		0.4	(3)
Gradient conditions			
B (per cent)	Time (min)	Time (min)	
30	0	0	
30	3	$(3 \times 0.4) = 1.2$	
70	13	$[1.2 + (10 \times 0.4)] = 5.2$	
30	16	$[5.2 + (3 \times 0.4)] = 6.4$	

(1) 11 per cent increase within allowed L/dp change of – 25 per cent to + 50 per cent;

(2) calculated using $F_2 = F_1 [(dc_2^2 \times dp_1) / (dc_1^2 \times dp_2)]$;

(3) calculated using $t_{G2} = t_{G1} \times (F_1 / F_2) [(L_2 \times dc_2^2) / (L_1 \times dc_1^2)]$.

Column temperature: $\pm 5^\circ C$, where the operating temperature is specified, unless otherwise prescribed.

Further adjustments in analytical procedure conditions (mobile phase, temperature, pH, etc.) may be required within the permitted ranges described in this chapter under System suitability and Adjustment of chromatographic conditions.

Mobile phase

Composition/gradient: adjustments of the composition of the mobile phase and the gradient are acceptable provided that:

- the system suitability requirements are fulfilled;
- the principal peak(s) elute(s) within ± 15 per cent of the indicated retention time(s) obtained with the original conditions; this requirement does not apply when the column dimensions are changed;
- the composition of the mobile phase and the gradient are such that the first peaks are sufficiently retained and the last peaks are eluted.

pH of the aqueous component of the mobile

phase: ± 0.2 pH units, unless otherwise prescribed.

Concentration of salts in the buffer component of a mobile phase: ± 10 per cent.

Where compliance with the system suitability requirements cannot be achieved, it is often preferable to reconsider the dwell volume or to change the column.

Dwell volume

The configuration of the equipment employed may significantly alter the resolution, retention time and relative retentions described. Should this occur, it may be due to a change in dwell volume. Monographs preferably include an isocratic step before the start of the gradient programme so that an adaptation can be made to the gradient time points to take account of differences in dwell volume between the system used for analytical procedure development and that actually used. It is the user's responsibility to adapt the length

of the isocratic step to the analytical equipment used. If the dwell volume used during the elaboration of the monograph is given with the monograph, the time points (t), in minutes, stated in the gradient table may be replaced by adapted time points (t_c), in minutes, calculated using the following equation:

$$t_c = t - \frac{(D - D_0)}{F}$$

- D = dwell volume, in millilitres;
 D_0 = dwell volume used for development of the analytical procedure, in millilitres;
 F = flow rate, in millilitres per minute.

The isocratic step introduced for this purpose may be omitted if validation data for application of the analytical procedure without this step is available.

Detector wavelength: no adjustment permitted.

Injection volume: when the column dimensions are changed, the following equation may be used for adjusting the injection volume:

$$V_{inj2} = V_{inj1} \times \frac{L_2 \times dc_1^2}{L_1 \times dc_2^2}$$

- V_{inj1} = injection volume indicated in the monograph, in microlitres;
 V_{inj2} = adjusted injection volume, in microlitres;
 L_1 = column length indicated in the monograph, in millimetres;
 L_2 = new column length, in millimetres;
 dc_1 = column internal diameter indicated in the monograph, in millimetres;
 dc_2 = new column internal diameter, in millimetres.

This equation may not be applicable to changes from TPP columns to SPP columns.

Even in the absence of any column dimension change, the injection volume may be varied provided the system suitability criteria remain within their established acceptability limits. When the injection volume is decreased, special attention is given to (limit of) detection and repeatability of the peak response(s) to be determined. An increase is permitted provided that, in particular, linearity and resolution of the peak(s) to be determined remain satisfactory.

GAS CHROMATOGRAPHY

Column parameters

Stationary phase:

- *particle size*: maximum reduction of 50 per cent; no increase permitted (packed columns);
- *film thickness*: – 50 per cent to + 100 per cent (capillary columns).

Column dimensions:

- *length*: – 70 per cent to + 100 per cent;
- *internal diameter*: ± 50 per cent.

Column temperature: ± 10 per cent.

Temperature programme: adjustment of temperature is permitted as stated above; adjustment of ramp rates and hold times of up to ± 20 per cent is permitted.

Flow rate: ± 50 per cent.

The above changes are acceptable provided the system suitability requirements are fulfilled and the selectivity and elution order of the specified impurities to be controlled are demonstrated to be equivalent.

Injection volume and split ratio: may be varied provided the system suitability criteria remain within their established acceptability limits. When the injection volume is decreased or the split ratio is increased, special attention is given to

(limit of) detection and repeatability of the peak response(s) to be determined. An increase in injection volume or a decrease in split ratio is permitted provided that, in particular, linearity and resolution of the peak(s) to be determined remain satisfactory.

Injection port temperature and transfer-line temperature in static head-space conditions: ± 10 °C, provided no decomposition or condensation occurs.

QUANTITATION

The following quantitation approaches may be used in general texts or monographs.

External standard method

Using a calibration function. Reference solutions with several graded amounts of a reference standard of the component to be analysed are prepared in a range that has been demonstrated to give a linear response, and a fixed volume of these reference solutions is injected. With the chromatograms obtained, a calibration curve is drawn by plotting the peak areas or peak heights on the ordinate against the amount of reference standard on the abscissa. The calibration function is generally obtained by linear regression. Then a solution of the substance to be examined is prepared according to the procedure specified in the individual monograph. The chromatography is performed under the same operating conditions as for the preparation of the calibration function. The peak area or peak height of the component to be analysed is measured, and the amount of the component is read or calculated from the calibration function.

Using one-point calibration. In an individual monograph, generally one of the reference solutions with a concentration within the linear range of the calibration function and a solution of the substance to be examined with a concentration close to that of the reference solution are prepared. The chromatography is performed under fixed conditions to obtain the amount of the component to be analysed by comparing the responses obtained. In this method, all procedures, such as the injection procedure, must be carried out under constant conditions.

Internal standard method

Using a calibration function. In the internal standard method, a stable component that shows a retention time close to that of the component to be analysed, and whose peak is well separated from all other peaks in the chromatogram, is chosen as an internal standard. Several reference solutions containing a fixed amount of the internal standard and several graded amounts of a reference standard of the component to be analysed are prepared. Based on the chromatograms obtained by injection of a fixed volume of individual reference solutions, the ratio of peak area or peak height of the reference standard to that of the internal standard is calculated. A calibration curve is drawn by plotting these ratios on the ordinate against the amount of reference standard (or the ratio of the amount of reference standard to that of the internal standard), on the abscissa. The calibration function is generally obtained by linear regression. Then a solution of the substance to be examined containing the same amount of internal standard as in the reference solutions used for the preparation of the calibration function is prepared according to the procedure specified in the individual monograph. The chromatography is performed under the same operating conditions as for the preparation of the calibration function. The ratio of the peak area or peak height of the component to be analysed to that of the internal standard is calculated, and the amount of the component is read or calculated from the calibration function.

Using one-point calibration. In an individual monograph, generally one of the reference solutions with a concentration within the linear range of the calibration function and a solution of the substance to be examined with a concentration close to that of the reference solution (both containing a fixed amount of the internal standard) are prepared. The

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chromatography is performed under fixed conditions to determine the amount of the component to be analysed by comparing the ratios obtained.

Normalisation procedure

Provided linearity of the peaks has been demonstrated, individual monographs may prescribe that the percentage content of a component of the substance to be examined is calculated by determining the area of the corresponding peak as a percentage of the total area of all the peaks, excluding those due to solvents or reagents, those arising from the mobile phase or the sample matrix, and those at or below the reporting threshold.

OTHER CONSIDERATIONS

Detector response

The detector sensitivity is the signal output per unit concentration or unit mass of a substance in the mobile phase entering the detector. The relative detector response factor, commonly referred to as the response factor, expresses the sensitivity of a detector for a given substance relative to a reference substance. The correction factor is the reciprocal of the response factor. In tests for related substances, any correction factors indicated in the monograph are applied (i.e. when the response factor is outside the range 0.8-1.2).

Interfering peaks

Peaks due to solvents and reagents or arising from the mobile phase or the sample matrix are disregarded.

Measurement of peaks

Integration of the peak area of any impurity that is not completely separated from the principal peak is typically performed by tangential skim (Figure 2.2.46.-9).

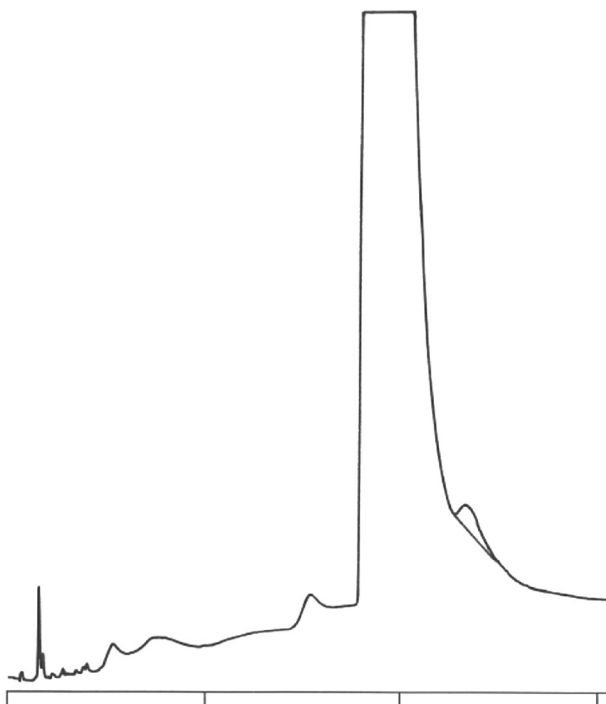


Figure 2.2.46.-9.

Reporting threshold

When the related substances test prescribes a limit for the total of impurities, or a quantitative determination of one impurity, it is important to choose an appropriate reporting threshold and appropriate conditions for the integration of the peak areas.

In such tests, the reporting threshold, i.e. the limit above which a peak is reported, is generally 0.05 per cent.



2.2.47. CAPILLARY ELECTROPHORESIS⁽⁸⁾

GENERAL PRINCIPLES

Capillary electrophoresis is a physical method of analysis based on the migration, inside a capillary, of charged analytes dissolved in an electrolyte solution, under the influence of a direct-current electric field.

The migration velocity of an analyte under an electric field of intensity E , is determined by the electrophoretic mobility of the analyte and the electro-osmotic mobility of the buffer inside the capillary. The electrophoretic mobility of a solute (μ_{ep}) depends on the characteristics of the solute (electric charge, molecular size and shape) and those of the buffer in which the migration takes place (type and ionic strength of the electrolyte, pH, viscosity and additives). The electrophoretic velocity (v_{ep}) of a solute, assuming a spherical shape, is given by the equation:

$$v_{ep} = \mu_{ep} \times E = \left(\frac{q}{6\pi\eta r} \right) \times \left(\frac{V}{L} \right)$$

- q = effective charge of the solute,
- η = viscosity of the electrolyte solution,
- r = Stoke's radius of the solute,
- V = applied voltage,
- L = total length of the capillary.

When an electric field is applied through the capillary filled with buffer, a flow of solvent is generated inside the capillary, called electro-osmotic flow. The velocity of the electro-osmotic flow depends on the electro-osmotic mobility (μ_{eo}) which in turn depends on the charge density on the capillary internal wall and the buffer characteristics. The electro-osmotic velocity (v_{eo}) is given by the equation:

$$v_{eo} = \mu_{eo} \times E = \left(\frac{\varepsilon\zeta}{\eta} \right) \times \left(\frac{V}{L} \right)$$

- ε = dielectric constant of the buffer,
- ζ = zeta potential of the capillary surface.

The velocity of the solute (v) is given by:

$$v = v_{ep} + v_{eo}$$

The electrophoretic mobility of the analyte and the electro-osmotic mobility may act in the same direction or in opposite directions, depending on the charge of the solute. In normal capillary electrophoresis, anions will migrate in the opposite direction to the electro-osmotic flow and their velocities will be smaller than the electro-osmotic velocity. Cations will migrate in the same direction as the electro-osmotic flow and their velocities will be greater than the electro-osmotic velocity. Under conditions in which there is a fast electro-osmotic velocity with respect to the electrophoretic velocity of the solutes, both cations and anions can be separated in the same run.

The time (t) taken by the solute to migrate the distance (l) from the injection end of the capillary to the detection point (capillary effective length) is given by the expression:

(8) This chapter has undergone pharmacopoeial harmonisation. See chapter 5.8. *Pharmacopoeial harmonisation*.

ACCURACY

Verify the accuracy preferably by using a certified reference material (CRM). Where this is not possible, perform a test for recovery.

Recovery. For assay determinations a recovery of 90 per cent to 110 per cent is to be obtained. The test is not valid if recovery, for example for trace-element determination, is outside the range 80 per cent to 120 per cent of the theoretical value. Recovery may be determined on a suitable reference solution (matrix solution) spiked with a known quantity of analyte (concentration range that is relevant to the samples to be determined).

REPEATABILITY

The repeatability is not greater than 3 per cent for an assay and not greater than 5 per cent for an impurity test.

LIMIT OF QUANTIFICATION

Verify that the limit of quantification (for example, determined using the 10 σ approach) is below the value to be measured.



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2.2.59. GLYCAN ANALYSIS OF GLYCOPROTEINS

1. INTRODUCTION

Glycan analysis is a test to analyse glycan moieties of glycoproteins. It may involve:

- whole glycoprotein analysis;
- separation and detection of protein glycoforms;
- analysis of glycopeptides obtained after enzymatic treatment of the glycoprotein;
- analysis of released glycans obtained after chemical or enzymatic treatment of the glycoprotein.

Monosaccharide analysis may complement information obtained by glycan analysis.

Glycosylation can play a predominant role in determining the function, pharmacokinetics, pharmacodynamics, stability, and immunogenicity of biotherapeutics. Glycosylation, unlike transcription, is a non-template-driven enzymatic modification process that results in glycan heterogeneity. The manufacturing procedure also has an influence on glycan heterogeneity. Glycoprotein glycan analysis may therefore be an important test to identify variations in the glycosylation pattern of the glycoprotein and/or monitor the consistency of the glycosylation pattern during production.

Glycan analysis can be a comparative procedure, because the information obtained, compared to a similarly treated reference substance, confirms product consistency.

This chapter provides approaches used for glycoprotein glycan analysis and requirements for the application of methods and validation of methods.

Glycan analysis is not a single general method, but involves the application of specific procedures and the development of specific glycan maps for each unique glycoprotein. Specific procedures are therefore indicated in relevant specific monographs.

1-1. PROTEIN GLYCOSYLATION

There are 3 main types of enzymatic glycosylation found in proteins:

- *N*-glycosylation, which involves the addition of oligosaccharides to the nitrogen on the terminal amide group of asparagine;

- *O*-glycosylation, which involves the addition of oligosaccharides to the hydroxyl groups of serine, threonine, and/or hydroxyproline;
- *C*-glycosylation, which involves the addition of an α -mannopyranose to the C2-carbon of the indole ring of tryptophan.

Non-enzymatic additions, also known as glycation, can occur when proteins are incubated with reducing sugars.

This chapter describes analytical methods for the *N*- and *O*-linked glycosylations, which are the most commonly found in glycoprotein medicinal products.

1-2. HETEROGENEITY OF THE PROTEIN GLYCOSYLATION

Different levels of glycan heterogeneity can appear during the production of glycoproteins. This heterogeneity may result from variations:

- in the degree of occupancy (full, partial, unoccupied);
- in the type of glycosylation (*N*- or *O*-linked);
- in the oligosaccharide structures (extensions, branching and linkage).

This heterogeneity in glycosylation results in a set of glycoforms for one specific glycoprotein. These variations arise because, unlike transcription and translation, glycosylation is a non-template post-translational modification process. The glycosylation pattern at a given site depends on many factors including the cell-specific and/or growth-dependent availability of glycosyltransferases and exo-glycosidases found in the Golgi apparatus and endoplasmic reticulum. Protein glycosylation is also influenced by the protein structure, the production process, the host-vector expression system and the cell culture conditions.

2. GLYCAN ANALYSIS PROCEDURES

Heterogeneity in glycosylation can be assessed by 4 distinct and complementary approaches:

- analysis of the intact glycoprotein;
- analysis of glycopeptides;
- analysis of released glycans;
- monosaccharide analysis.

The present section provides methods and general requirements used for glycan analysis of glycoproteins containing *N*- and *O*-linked glycans.

Glycan analysis is usually a multistep process. There are numerous methodologies for glycan analysis. This variety is a consequence of the diversity and complexity of glycan structures, of the available technologies and detection systems, and of the wide range of approaches depending on the level of information required.

Figure 2.2.59.-1 provides an overview of glycan analysis analytical procedures that can be employed to apply the chosen approach(es). Many variations of the same techniques and conditions are available depending on the glycan structures and origin.

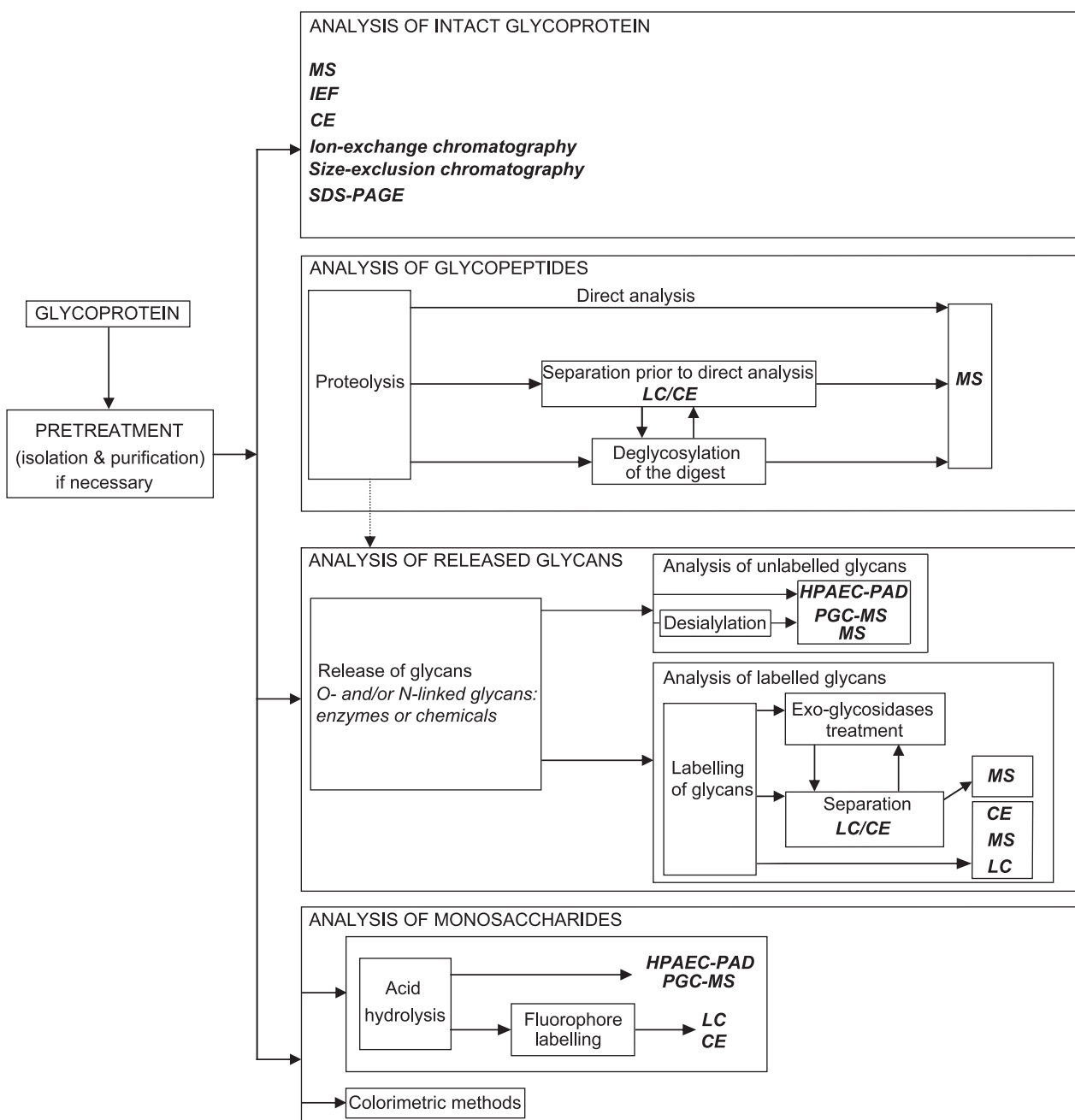
Isolation and purification. Isolation and purification may be necessary for analysis of bulk drug substances or dosage forms containing interfering excipients, and, when required, will be described in the specific monograph.

2-1. ANALYSIS OF INTACT GLYCOPROTEIN

Analysis of the intact glycoprotein provides information on the overall pattern of glycosylation of the glycoprotein.

This approach provides limited information when the molecule is large and contains multiple glycosylation sites.

Methods such as capillary electrophoresis (CE) (2.2.47) and mass spectrometry (MS) (2.2.43) can be used. Size-based techniques, such as size-exclusion chromatography (2.2.30) and sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) (2.2.31), may provide information on the glycosylation status of a protein. If the degree of sialylation



CE: Capillary electrophoresis

HPAEC-PAD: High-pH anion-exchange chromatography with pulsed amperometric detection

IEF: Isoelectric focusing

LC: Liquid chromatography

MS: Mass spectrometry

PGC: Porous graphite chromatography

SDS-PAGE: Sodium dodecyl sulfate polyacrylamide gel electrophoresis

Figure 2.2.59.-1. – Overview of glycan analysis procedures

significantly contributes to the biological activity of the glycoprotein, ion-exchange chromatography (2.2.46), isoelectric focusing (IEF) (2.2.54) or CE (2.2.47) may be performed to monitor sialylation. The technique must be chosen according to its suitability to provide a reliable correlation between the degree of sialylation and the bioactivity of the product.

2-2. ANALYSIS OF GLYCOPEPTIDES

Analysis of glycopeptides provides information on site-specific glycosylation properties, on the degree of occupancy, and on the oligosaccharide structures. It involves proteolytic digestion of the glycoprotein. Approaches to site-specific cleavage of the protein backbone are given in general chapter 2.2.55. *Peptide mapping*.

After proteolysis of the glycoprotein, the following approaches can be chosen.

Direct analysis by MS (2.2.43). Care should be taken that the glycopeptide signal is not suppressed due to the presence of other peptides, where glycopeptides represent a minor portion of the total peptide mixture and where signal intensities are lower than those of non-glycosylated peptides.

Separation prior to analysis by MS. This additional step overcomes the problems raised above. Enrichment or fractionation techniques can be used either in parallel with or sequentially to direct analysis. Separation techniques such as liquid chromatography (LC) (2.2.29) and CE (2.2.47) are suitable. These techniques may be interfaced with MS to allow online MS measurements.

Deglycosylation of the glycopeptides. Identification of the different glycosylation sites of a glycoprotein is made possible by comparing peptide maps obtained by proteolytic digestion of the intact glycoprotein to those obtained when the glycoprotein is deglycosylated

previously or following proteolytic digestion. The peptide mass gives information about the glycosylation sites and by calculating the mass difference between the intact glycopeptide and the deglycosylated glycopeptide, it is possible to obtain information about the attached glycans concerning composition and heterogeneity. Approaches to deglycosylation of the protein backbone are given in section 2-3-1. A separation step can be performed after or before deglycosylation.

2-3. ANALYSIS OF RELEASED GLYCANS

Analysis of released glycans provides a convenient way to obtain information on the various populations of glycans present on the protein (bi-, tri-, and tetra-antennary profile). The degree of sialylation can also be addressed at this stage. Depending on the chosen method, prior derivatisation/labelling may be needed to allow the detection of the glycans.

Analysis of released glycans generally involves the release and purification of glycans from the reaction mixture, followed by the labelling/derivatisation of the glycans, where needed; the glycans are then profiled (fractionation or separation).

2-3-1. Release of glycans

The selection of the approach used for the release of glycans will depend on the glycoprotein under test. The cleavage agent to be employed is chosen according to the type of cleavage needed and level of information required. Enzymatic or chemical cleavage may be used. Table 2.2.59.-1 gives a non-exhaustive list of enzymatic cleavage agents and their specificity.

Digestion efficiency is generally dependent on the accessibility of the glycans on the protein and hence the protein can be denatured to maximise glycosylation site exposure, unless it is desirable to distinguish between surface and buried glycans.

Chemical cleavage agents might also be used, using for example hydrazine or alkaline borohydride for β -elimination.

Table 2.2.59.-1. – Examples of enzymatic cleavage agents

Agents	Specificity
N-linked glycans release	
Peptide-N⁴-(N-acetyl-β-glucosaminyl)asparagine amidase (EC 3.5.1.52)	Hydrolysis of an N⁴-(acetyl-β-D-glucosaminyl)asparagine residue in which the glucosamine residue may be further glycosylated, to yield a (substituted) N-acetyl-β-D-glucosaminylamine and a peptide containing an aspartate residue
- Peptide N-glycosidase F (PNGase F)	Release of N-glycan chain but no release of N-glycan chain containing (α 1-3)-linked core fucose
- Peptide N-glycosidase A (PNGase A)	Release of N-glycan chain containing (α 1-3)-linked core fucose
Mannosyl-glycoprotein endo-β-N-acetylglucosaminidase (EC 3.2.1.96)	Endohydrolysis of the N,N'-diacetylchitobiosyl unit in high-mannose glycopeptides/glycoproteins containing the -[Man(GlcNAc)₂]Asn structure
- Endo- β -N-acetylglucosaminidase F (endo F)	Release of high-mannose, hybrid and complex oligosaccharides
- Endo- β -N-acetylglucosaminidase H (endo H)	Release of high-mannose, hybrid oligosaccharides
O-linked glycans release	
Glycopeptide α-N-acetylgalactosaminidase (EC 3.2.1.97)*	Hydrolysis of terminal D-galactosyl-N-acetyl-α-D-galactosaminidic residues
* This enzyme has limited usage because of its high substrate specificity.	

2-3-2. Analysis of glycans

Released glycans can be analysed or profiled by chromatographic, electrophoretic and mass spectrometric techniques, and in general by a combination of these techniques. The choice of the method can be grouped according to the nature of the glycans and level of information required.

Analysis of glycans provides information on the various populations of glycans present on the protein (high-mannose, hybrid, complex). Information on the relative amounts of branched structures might be obtained by analysis of desialylated glycans.

A separation step may be required. It implies the use of LC (2.2.29) and CE (2.2.47) as intermediate techniques. LC (2.2.29) can be used preparatively with individual fractions being collected (usually labelling is required) or can be directly coupled to MS (2.2.43).

2-3-2-1. Analysis of unlabelled glycans

Native glycans can be analysed by high-pH anion-exchange chromatography with pulsed amperometric detection (HPAEC-PAD), porous graphite chromatography (PGC) and MS (2.2.43).

HPAEC-PAD has high sensitivity and can also separate some linkage isomers. Response factors of the different signals are not equal for the different oligosaccharide structures. Absolute quantification of the glycan is not possible unless an oligosaccharide reference library is available. Quantification can be obtained by comparison with a well-characterised reference standard of the substance being tested, or by relating the peak area of each glycan to the total peak area of all glycans in the map.

PGC can also be used to separate native glycans because of its higher selectivity compared to the conventional non-polar columns. A PGC-electrospray-ionisation-MS approach can be applied for direct glycan analysis.

2-3-2-2. Analysis of labelled glycans

Labelling of glycans

The type of derivatisation carried out will depend on the method used to detect glycans: UV or fluorescent.

Derivatisation with fluorescent labels is the most commonly used technique for labelling glycans at their reducing end by reductive amination. One label can be attached to every single mono- and oligo-saccharide, allowing the determination of molar quantities. Table 2.2.59.-2 gives a non-exhaustive list of commonly used fluorescent labels and suitable analytical techniques.

Table 2.2.59.-2. – Examples of fluorescent labels and suitable techniques

Name	Acronym	Analytical techniques
2-Aminobenzoic acid	2-AA	LC (2.2.29), MS (2.2.43)
2-Aminobenzamide	2-AB	LC (2.2.29), MS (2.2.43)
2-Aminopyridine	2-AP	LC (2.2.29), MS (2.2.43)
2-Amino-9(10H)-acridinone	AMAC	Gel electrophoresis (2.2.31)
Trisodium 8-aminopyrene-1,3,6-trisulfonic acid	APTS	CE (2.2.47)

Permethylation of glycans may also be used when MS (2.2.43) is used alone for detection. It is based on the methylation of the oligosaccharides.

Analysis of labelled glycans

Labelled glycans can be analysed by analytical techniques such as LC (2.2.29), CE (2.2.47) and MS (2.2.43).

According to the separation properties of the glycans, glycans can be profiled and quantified by several LC (2.2.29) systems using an appropriate label: reversed-phase (separation by hydrophobicity), normal-phase (separation by size), and anion-exchange (separation by charge) LC.

2-4. MONOSACCHARIDE ANALYSIS

Monosaccharide analysis provides information on the monosaccharide composition of a glycoprotein. Analysis of monosaccharides can be performed using either colorimetric or separation methods.

2-4-1. Colorimetric methods

The colorimetric methods, which are based on chemical staining, provide information on the quantity of specific classes of sugars such as sialic acids, neutral sugars and hexosamines.

2-4-2. Separation methods

The separation methods generate quantitative information on the overall monosaccharide composition. The methods require acid hydrolysis pre-treatment of the oligosaccharide chains of the intact glycoprotein or released glycans, prior to analysis. To release sialic acids, mild acid hydrolysis or enzymatic treatment is employed. The hydrolysis step is a significant source of variability and may require product-specific validation.

Methods for separation and quantification of monosaccharides include:

- the use of HPAEC-PAD and PGC-MS, which allow the determination of molar quantities of native monosaccharides (sialic acids, neutral sugars and alcohol sugars);
- fluorophore labelling of monosaccharides followed by separation methods such as reversed-phase or ion-exchange chromatography, or CE.

3. EVALUATION AND ANALYSIS OF DATA

Data obtained from analytical methods for the analysis of glycans can be analysed and evaluated for 3 different purposes:

- confirmation of identity of individual structures or families of structures;
- confirmation of compliance of the substance being tested with qualitative requirements;
- confirmation of compliance of the substance being tested with quantitative requirements.

Specific considerations with respect to reference standards and method development of each level of analysis are set out in sections 4 and 5 respectively.

3-1. CONFIRMATION OF IDENTITY OF INDIVIDUAL STRUCTURES OR FAMILIES OF STRUCTURES

The analytical target for a glycan analysis method may be an individual monosaccharide (e.g. sialic acid, fucose), a defined oligosaccharide structure (e.g. tetra-sialylated, tetra-antennary glycan) or a family of structures sharing a common analytical feature (e.g. tetra-sialylated glycans, tri-antennary glycans, glycoprotein isoforms with the same charge). Confirmation of the identity of the analytical target is an essential step in the

analysis and evaluation of data, and can be achieved absolutely, by verification of molecular structure, or comparatively, by comparison with an appropriate reference standard.

3-1-1. Absolute confirmation of identity

Absolute confirmation of the identity of glycan structures is typically achieved during product development, and should not necessarily be the target of routine analysis. Identity of the analytical target will be assigned by reference to a known molecular property of the molecule. Such absolute identification of individual structures can require multi-step approaches using enzymatic and chemical reactions, separation techniques and online or offline detection methods, and will most commonly use the charge-to-mass ratio of a molecular ion, determined using a suitable mass spectrometric method as the final basis for structure assignment.

3-1-2. Comparative confirmation of identity

During routine application of the analytical method, the identity of the analytical target may be confirmed by comparison with process or system suitability reference standards. These may be generated from known, well-characterised glycoproteins, which may be of the same general class as the product being tested (e.g. fetuin for complex N-linked glycoproteins), or may be derived from a well-characterised batch of the product being tested, which has been established as a reference standard. The following considerations apply to comparative assignment of structural identity:

- in the case of a validated high reproducibility of the retention times, the absolute retention times can be used for correct assignment;
- alternatively, a glycan marker can be injected at the beginning and at the end of the testing sequence and checked for any drifts in the retention times; based on these reference chromatograms the glycans of the test samples can be assigned;
- in cases where no standard is available to assign all glycan peaks in the test sample, absolute or normalised retention times can be used to monitor and label unidentified glycan peaks.

3-2. CONFIRMATION OF COMPLIANCE OF THE SUBSTANCE BEING TESTED WITH QUALITATIVE REQUIREMENTS

At this level of evaluation, the analytical results obtained with the product being tested are evaluated to demonstrate compliance with specifications. Typically this is achieved by comparison with data obtained in parallel using a reference standard of the substance being tested. In evaluating the data it is necessary:

- to establish that the analytical result obtained using the reference standard is broadly comparable to the expected result, to verify the suitability of the system; for example, in a glycan mapping procedure, this would be achieved by comparison of the map obtained with the reference substance with a provided specimen map obtained during establishment of the reference substance, and by ensuring compliance with all stated system suitability criteria;
- to demonstrate similarity of the maps obtained with the reference substance and the test substance, using any specific compliance criteria given in the specific monograph.

3-3. CONFIRMATION OF COMPLIANCE OF THE SUBSTANCE BEING TESTED WITH QUANTITATIVE REQUIREMENTS

3-3-1. Quantitative measurement of analyte levels and expression of results

In some cases, e.g. measurement of sialic acid or other monosaccharides, data can be expressed in order to obtain a molar ratio of sialic acid to glycoprotein. Data is calculated by reference to a reference standard for sialic acid and to a validated method of protein determination. Either the internal or external standard method may be used (see general chapter 2.2.46. *Chromatographic separation techniques*).

3-3-2. Quantitative expressions of separation profile

Profiles or distribution patterns may be expressed numerically in a number of ways, including the normalisation procedure; the percentage content of each analytical target, e.g. glycan entity, is calculated by determining the response of the glycan entity as a percentage of the total response of all the entities, excluding those due to solvents or any added reagents, and those below the disregard limit. In addition, numerical expressions such as the Z number, which are method- and product-specific and defined in specific monographs, can be used.

4. REFERENCE STANDARDS

Reference standards for glycan analysis serve 2 functions: the verification of the suitability of the system and the confirmation that the article under test complies with specified requirements.

The reference standards used for system suitability may be:

- a reference substance for the substance being tested;
- glycan moieties liberated from a fully characterised reference standard of the substance being tested;
- well-characterised glycan moieties liberated from glycoproteins (e.g. fetuin, IgG);
- glycan markers characterised for identity and purity.

The reference standard used for compliance of the glycoprotein under test is a preparation of the substances being tested. It is noted that glycan analysis procedures described in specific monographs prescribe the use of a reference standard for the substance being tested and for which the glycan analysis procedure has been validated.

5. POINTS TO CONSIDER IN METHOD DEVELOPMENT

This section provides means for measuring the overall performance of the method during development. The extent of method development and analytical validation is selected on the basis of their suitability for a specific product. Depending on the chosen approach, several steps are necessary for glycan analysis, for example:

- isolation and purification (or desalting) of the glycoprotein;
- enzymatic (or chemical) treatment of the glycoprotein to selectively release either *N*- or *O*-linked glycans from the protein backbone;
- isolation and purification of the released glycans;
- verification of released sialic acid and monosaccharide residues;
- chromophore labelling of the released glycans;
- separation of the glycans, native or fluorescence labelled;
- glycan identification and quantification (e.g. determination of the Z number);
- determination of site occupancy based on relative quantities of glycosylated and non-glycosylated peptides.

Protein isolation and purification. Isolation and purification of the glycoprotein from its matrix may be necessary to remove all interfering substances (e.g. excipients, salts) and, when required, will be specified in the specific monograph. This must be performed in a reproducible manner in order to guarantee a quantitative recovery of the protein.

Release and isolation of oligosaccharides. The approach chosen for the release of glycans will depend on the protein under test and will be based on the types of glycosylation, i.e. *N*- or *O*-linked glycosylation. Non-compendial approaches available for the release of glycans must be optimised in order to ascertain a quantitative profiling of all glycan entities. Factors that impact cleavage efficiency, such as enzyme-to-protein concentration ratio, temperature, reaction time course, and denaturation of protein prior to digestion, must be optimised.

It is noteworthy that the enzymatic/chemical reaction must not alter the glycan composition, e.g. not destroy sialic acid residues. Where there is more than one glycosylation site, the enzymatic treatment should proportionally release all oligosaccharide moieties attached to the protein, independent of their structure and their individual position in the protein. Reproducible recovery of all glycan entities from the reaction mixture must be confirmed.

Derivatisation of released glycans. Derivatisation is usually carried out according to non-compendial protocols. Therefore, the reproducible derivatisation of all glycan entities must be verified. This may be achieved through optimisation of the reaction conditions such as amount of the derivatisation reagent, reaction temperature and time. The derivatisation reaction must not change the glycan composition, e.g. not destroy sialic acid residues.

Separation, identification and system suitability. The methods employed for glycan analysis must be capable of detecting and separating different glycan moieties to ascertain a reliable identification and quantification.

The acceptance criteria for system suitability, which also cover glycan cleavage, recovery and analysis, depend on the critical test parameters that affect the outcome of the result.

A comparison between the glycan map of the substance under test and that of a reference substance, being treated in the same conditions, is an indicator to evaluate the performance of the analytical procedure. In order to further confirm the obtained results, the analyses may be repeated with an orthogonal method. The use of a reference standard (e.g. reference substance of the product being examined, system suitability glycan marker) is essential in the establishment of system suitability parameters and validation of the analytical procedure.

Reproducibility of quantitative expression (e.g. Z number estimation) of glycan profiles must be verified.

Determination of site occupancy based on relative quantities of glycosylated and non-glycosylated peptides.

Where site occupancy is estimated by comparison of glycosylated and non-glycosylated peptides from an enzymatically digested glycoprotein, reproducible cleavage of both forms of the peptide must be demonstrated.

6. GLYCAN ANALYSIS DECISION-MAKING FRAMEWORK

This decision-making framework is given for information and does not constitute a mandatory part of the European Pharmacopoeia.

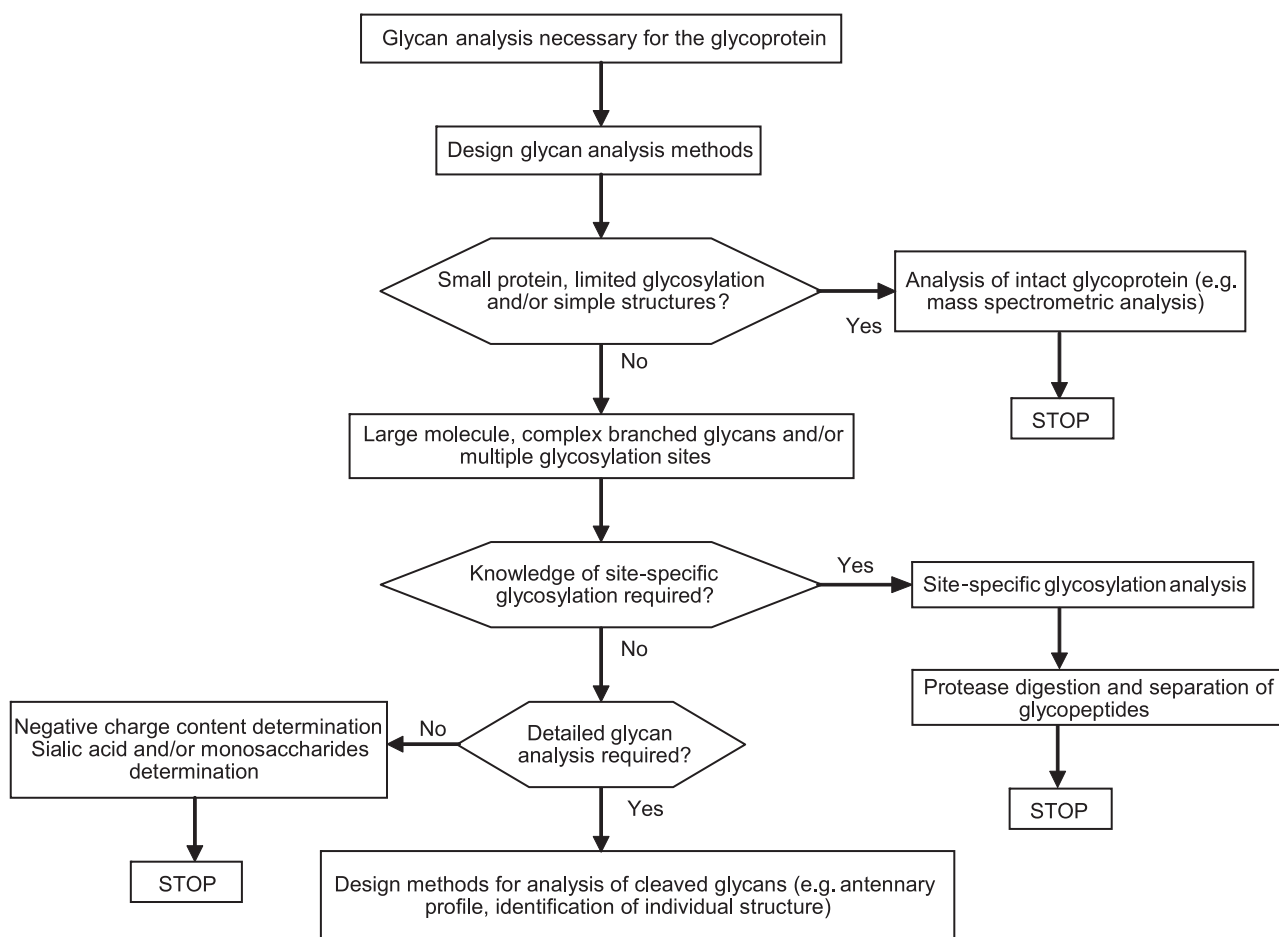


Figure 2.2.59.-2. – Guidance on methods to be used when glycan analysis is required

The choice of procedures used to analyse glycans is established according to the level of information required to ensure the quality of the glycoprotein and is set up during the development phase of the product.

Figure 2.2.59.-2 provides guidance in the choice of methods to be used when glycan analysis is required.

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2.2.61. CHARACTERISATION OF CRYSTALLINE SOLIDS BY MICROCALORIMETRY AND SOLUTION CALORIMETRY

For the purpose of this chapter, crystalline material, partially crystalline material and amorphous material are considered as solids.

INTRODUCTION - THE CONCEPT OF CRYSTALLINITY

The perfectly ordered crystal lattice with every molecule in its expected lattice position is an ideal that is seldom, if ever, achieved. The other extreme is the amorphous state, in which a solid contains the maximum possible density of imperfections (defects of various dimensionalities), such that all long-range order is lost while only the short-range order, imposed by its nearest neighbours, remains. Real crystals lie somewhere between these 2 extremes. A crystal's position on a scale bounded by these 2 extremes is termed *crystallinity*. All real crystals, even in the pure state, possess some lattice imperfections or defects, which increase both the energy (enthalpy under conditions of constant atmospheric pressure)

and the disorder (expressed as the entropy) of the crystal lattice. A crystal with a relatively low density of imperfections is said to be highly crystalline and to possess a high crystallinity. By contrast, a particle with a relatively high density of imperfections is said to be partially amorphous and to possess a low crystallinity. In ideal terms, a totally amorphous particle corresponds to zero crystallinity. Amorphous particles may contain somewhat ordered domains that can act as nuclei for crystallisation; such so-called amorphous particles are said to possess a low-level but finite crystallinity.

The ability to detect and to quantify the amount of amorphous material within a highly crystalline substance is of great importance during the development and subsequent manufacture of a pharmaceutical preparation.

In reality, a powder probably contains particles with different degrees of crystallinity, just as it may contain particles with varying sizes and shapes. The lower the crystallinity of a solid, the greater its enthalpy and entropy. The increase in enthalpy is never totally compensated for by the increase in entropy; therefore, the Gibbs free energy, which reflects the balance between them, actually increases. Hence, the lower the crystallinity of a material (powder), and consequently the greater its amorphous character, the greater its apparent intrinsic solubility and dissolution rate, but the lower its thermodynamic stability. Because of the great relevance of these properties, crystallinity is also an important property and requires measurement by a suitable method.

In the following chapter, the crystallinity or the content of amorphous parts of a powder are measured by calorimetric methods such as microcalorimetry or solution calorimetry, although other methods could be used (e.g. see general chapter 2.9.33. *Characterisation of crystalline and partially crystalline solids by X-ray powder diffraction (XRPD)*).